A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Module-by-module domain and level mapping, objective design and usage guide for AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semester II

Curriculum	NCISM, Board of Ayurveda — applicable from the 2024–25 batch
Coverage	8 modules · 42 units · 171 session learning objectives · 47 practical training and 44 experiential learning activities
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column 3E

171
OBJECTIVES CODED

105
THINKING

65
DOING

16
CREDITS · 480 HOURS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

Contents

PART A — UNDERSTANDING THE TAXONOMY

Chapter 1 What Bloom's taxonomy is, and why this book carries two versions 5

Chapter 2 The three domains 6

Chapter 3 The second axis 7

Chapter 4 Where Bloom already sits in your curriculum 8

Chapter 5 Verbs 9

Chapter 6 The one-to-two-to-three split 10

Chapter 7 Choosing how to teach, level by level 11

Chapter 8 Choosing how to check, level by level, and the modular assessment 12

Chapter 9 How to use a module and a unit in Part B 13

PART B — THE MODULES AND UNITS

Module 1 Pharmaceutics and Pharmaceutical Technology 24 obj · 56 h 15

Unit 1 Introduction to Pharmaceutics 6 obj

Unit 2 Pharmaceutical Dosage Forms 4 obj

Unit 3 Pharmaceutical Formulation 5 obj

Unit 4 Pharmaceutical Manufacturing and Quality Control 5 obj

Unit 5 Biopharmaceutics 4 obj

Module 2 Databases and Information Systems for Metal, Mineral, and Plant Research 22 obj · 56 h 17

Unit 1 Introduction to Databases and Information Systems 5 obj

Unit 2 Metal and Mineral Databases 6 obj

Unit 3 Plant Databases 4 obj

Unit 4 Database application 4 obj

Unit 5 Challenges and limitation of Databases 3 obj

Module 3 Essential Chemistry :- Unlocking the Rasashastra Techniques 21 obj · 56 h 19

Unit 1 Atomic, Molecular Structure, and Periodic Table Elements 3 obj

Unit 2 Specialized Chemistry Fields-An Outline 2 obj

Unit 3 Organic Chemistry 3 obj

Unit 4 Phytochemistry 7 obj

Unit 5 Metallurgy 6 obj

Module 4 Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures 15 obj · 56 h 21

Unit 1 Shodhana and Marana Procedures 3 obj

Unit 2 SOP of Shodhana Processes 4 obj

Unit 3 SOP of Marana process 4 obj

Unit 4 Chemical Changes involved in Shodhana and Marana processes 1 obj

Unit 5 Shodhana and Marana- Case Studies, troubleshooting and emerging trends 3 obj

Module 5 Principles, specification, SOPs of handling of classical and advanced instruments 29 obj · 84 h 23

Unit 1 Puta and its modern aspects 4 obj

Unit 2 Musha and its modern aspects 4 obj

Unit 3 Chromatography 5 obj

Unit 4 Spectroscopy 4 obj

Unit 5 Microscopy 4 obj
Unit 6 Calibration techniques 3 obj
Unit 7 Overview of characterization instruments 5 obj
Module 6 Animal house and Animal handling technique 13 obj · 28 h 25
Unit 1 Introduction to animal house management 3 obj
Unit 2 Study of animal behaviour 3 obj
Unit 3 Species-specific handling techniques 2 obj
Unit 4 Animal health and disease prevention 3 obj
Unit 5 Animal welfare and ethics 2 obj
Module 7 Classical and Advanced quality control techniques for Ayurveda Formulations 26 obj · 56 h 27
Unit 1 Quality Control and Quality Assurance 4 obj
Unit 2 Classical aspects of the quality control parameter for Ayurvedic formulations 4 obj
Unit 3 Quality Control parameter for solid dosage forms 9 obj
Unit 4 Quality Control parameter for semi-solid dosage forms 3 obj
Unit 5 Quality Control parameter for liquid and gaseous dosage forms 6 obj
Module 8 Nano-science, nano-technology and its application in Ayurveda 21 obj · 56 h 29
Unit 1 Introduction to Nanotechnology 3 obj
Unit 2 Ayurvedic Formulations and Nanotechnology 10 obj
Unit 3 Nanocarriers for Ayurvedic Formulations 3 obj
Unit 4 Regulatory Aspects and Safety Concerns 2 obj
Unit 5 Recent Advances and Future Directions 3 obj

PART C — REFERENCE

C1 The sixteen curriculum codes, decoded 31
C2 The two schemes side by side 33
C3 Level profile of every module and unit 35
C4 Verb index 37
C5 Kinds of knowledge across the semester 39
C6 All 47 practical training activities, mapped 41
C7 All 44 experiential learning activities, mapped 43
C8 Where the hours go, module by module 45
C9 Sources and further reading 47

PART A

Understanding the taxonomy

Nine short chapters. The first five explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, where it already sits inside this curriculum, and how verbs turn it into practice. Chapter 6 deals with something specific to postgraduate study here — the fixed one-to-two-to-three split between lecture, practical training and experiential learning, which decides more about what a student reaches than any other single feature of the course. The last three cover teaching, checking, and how to read Part B.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Column 3E
5. **Verbs** — the working end of the taxonomy
6. **The one-to-two-to-three split** — lecture, practical training, experiential learning
7. **Choosing how to teach**, level by level
8. **Choosing how to check**, level by level, and the modular assessment
9. **How to use a module and a unit in Part B**

↑ Contents

PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed what every teacher notices sooner or later: two questions on the same content can be worlds apart, not because one covers more ground, but because one asks the student to *do* something harder with what they know.

What they produced was a ladder of six levels, each asking more than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That the levels stack is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

- The levels became **verbs**: *Remember, Understand, Apply, Analyse, Evaluate, Create.*
- **The top two swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is 5, Create is 6, on the reasoning that making something new is a larger act than judging something that exists.
- A **second axis** was added for the kind of knowledge involved. Chapter 3 covers it.

> **The one thing to carry away.** When a colleague says a question is "at the top level", ask which scheme they mean. In 1956 the top is Evaluation — judging. In 2001 it is Create — producing. Different questions, different teaching, different marking. Most disagreement between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

Your curriculum uses the **1956 scheme**: the codes in Column 3E are CK, CC, CAP, CAN, CS and CE. If this book used only the 2001 names you could not match it against your syllabus document. But most faculty development now teaches the 2001 scheme. So every level here is named twice, 1956 first: *Comprehension / Understand.* Where the two schemes disagree about position — only at the top two — the book says so on the page.

> **This curriculum reaches the top, and that is unusual.** The undergraduate AyUG-RB syllabus does not use Synthesis at all. Semester II uses it four times, and it uses the top of the doing ladder twice. If you have taught only the undergraduate course, the ceiling here is genuinely higher and the teaching has to change accordingly.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee planned three volumes. The **cognitive** volume appeared in 1956; the **affective** volume — attitudes and values — in 1964; the **psychomotor** volume was never written by that committee, and the version your curriculum uses comes from later work by others.

THINKING — SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. In this semester, 105 of the 171 objectives are cognitive.

DOING — FIVE LEVELS

- **Set** — ready to begin; knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

Sixty-five of the 171 objectives are here — 38% of the semester. That is the single most important fact about this curriculum. It is not a course that can be delivered from the front of a room. The largest single group is *Mechanism*, and getting a student from guided to unaided is what most of the practical hours are for.

GOING ABOUT IT — FIVE LEVELS

Receiving, Responding, Valuing, Organisation, Characterisation — from paying attention when it matters, through treating something as worth doing unprompted, to it having become how the student works.

> **This semester is almost silent here.** One objective in 171 carries an affective code, in Module 2 Unit 5. That is worth knowing rather than working around: professional habits — verifying material, recording the reading actually taken, handling animals properly — are being built in this semester whether or not the curriculum codes them. If you want evidence of them, you will have to plan for it yourself, because the syllabus does not ask. Part B therefore shows two ladders per unit rather than three; the single affective objective appears in its own band in the objective table of the unit that holds it.

THE DOMAINS DO NOT MOVE TOGETHER

An objective at Mechanism may need only level 1 thinking if the procedure is followed by rote. Another at Set may need level 4 thinking if choosing the right equipment requires judgement. Module 3 in this semester reaches only Analysis on the thinking ladder but *Origination* on the doing ladder — the highest doing level in the whole course. Read both ladders.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a second question. The first is: what thinking must the student do? The second is: *what kind of knowledge is it about?*

- **Factual** — terminology, names, specific details. Learned by rehearsal and self-testing.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody.
- **Procedural** — methods, techniques, criteria for choosing between them. Learned by carrying them out.
- **Metacognitive** — knowing about your own learning. Learned by being made to look back at how you worked.

> **This semester is overwhelmingly Procedural.** Of the 171 objectives, 99 work on procedural knowledge, 70 on conceptual and only 10 on factual. That is the profile of a laboratory and workshop course rather than a reading course, and it lines up with the 38% psychomotor figure. The practical consequence is blunt: **reading a procedure is not learning it**, and a student who has only read will look identical to one who has practised right up until the moment they are asked to perform.

WHY IT CHANGES THE TEACHING

Two objectives at the same level can need entirely different sessions. One asks for the names in a group; another asks for the steps of a procedure in order. Rehearsal fixes the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession. This is why “they know it but they cannot do it” is such a common complaint — usually the objective was Procedural and the teaching was pitched as though it were Factual.

In Part B every objective is labelled with its kind of knowledge, read from the words of the objective itself. Where an objective works on two kinds, both are named. Where the wording supports a reading without settling it, the label carries a small circle (°), so you can see a judgement was made and overrule it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds. It has been in the curriculum since publication, in a column most teachers have never had pointed out.

COLUMN 3E

Open Table 3. The fifth column is headed *Domain/Sub Domain*. Every one of the 171 session learning objectives carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC through AFT-CHR for how the student goes about the work. The full key is Appendix C1.

THE COLUMNS AROUND IT

- **3A — Course outcome.** Which of the course-level outcomes the objective serves.
- **3C — Notional learning hours.** How long the objective is expected to take.
- **3D — Lecture, practical training or experiential learning.** The mode, and for the last two a reference into the activity tables. Chapter 6 is about this column.
- **3G — Teaching and learning methods.**

> **A word about Column 3F, which this book does not use.** Column 3F is headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It has its own companion volume. In *this* book, "level" always means the Bloom level in Column 3E. Where you read "level 5" here, it never means "Shows how".

MODULE LEARNING OBJECTIVES SIT ABOVE SESSION OBJECTIVES

Each module states its own learning objectives, and these are what the modular assessment tests. They are pitched at their own levels, which do not always match the session objectives beneath them. Part B shows each module's objectives with the level of their verb, so you can see at a glance whether the module assessment is aimed higher or lower than the sessions that feed it.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea; a verb is what a student meets. **An objective is taught and examined at the level of its verb**, whatever else is written beside it.

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, observe, record.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, differentiate, outline, present.
- **Level 3, Application / Apply** — apply, demonstrate, perform, prepare, calculate, measure, conduct, practise, operate, handle.
- **Level 4, Analysis / Analyse** — analyse, compare, examine, determine, categorise, correlate, review, explore, investigate, search.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, devise, synthesise, integrate, optimise. (Level 6, Create, in the 2001 scheme.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, interpret. (Level 5 in the 2001 scheme.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.*

> **Where a verb and a code disagree, follow the verb — and know you are choosing.** The verb is what appears on the paper, so it wins by default. That is fine provided it is a decision rather than an accident. If the verb asks less than the code, you have room to teach higher than the syllabus requires. If it asks more, plan for the verb. Part B tells you, objective by objective, which case you are in.

WRITING YOUR OWN

Decide the level first, then take a verb from that level, then check the verb can be assessed by the tool you have in mind. One verb, one level, one objective. Where you find yourself writing “identify, demonstrate and evaluate” on a single line, you have three objectives at three levels and they will need three different checks.

PART A · CHAPTER 6

The one-to-two-to-three split — lecture, practical training, experiential learning

This is the feature of postgraduate study that most decides what students actually reach, and it has no equivalent in the undergraduate course. Every credit in this semester is thirty hours, and the curriculum fixes how those hours are spent: **five hours of lecture, ten of practical training, thirteen of experiential learning, and two of modular assessment** — a ratio of roughly one to two to three.

Across the semester that comes to **80 lecture hours, 160 practical and 240 experiential**. Two thirds of the course happens with the teacher not talking.

THE THREE KINDS OF HOUR DO NOT REACH THE SAME LEVELS

- **Lecture** puts material in front of students and lets them retrieve it. It builds level 1 well and level 2 poorly, and on its own it builds nothing above level 2. Eighty of the 171 objectives are set as lecture.
- **Practical training** moves a student from performing with help to performing unaided. That single transition — Guided response to Mechanism — is what these hours are for, and it is slow. Forty-seven objectives are set here.
- **Experiential learning** hands over an unfamiliar situation and requires a judgement, a decision or a product. This is where the upper levels are actually built, and where the modular assessment lives. Forty-four objectives are set here.

> **The commonest failure is silent demotion.** An objective set as experiential learning and then delivered as a lecture has been quietly moved down two levels, and nothing in the timetable will show it. The students will have been told about the thing instead of doing it. Part B prints the kind of hour against every objective for this reason: it is the clearest statement the curriculum makes about how a thing is meant to be learned.

THE ACTIVITY TABLES

Every practical and experiential objective carries a reference — 1.1, 8.2 and so on — into the curriculum's activity tables, which describe what the student is to do. There are 47 practical training activities and 44 experiential ones. Appendices C6 and C7 reproduce all 91 with the level each serves, so you can see what a reference points at without turning back to the syllabus.

PART A · CHAPTER 7

Choosing how to teach, level by level

The commonest planning error is to choose the method by the hour rather than by the level. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. Make students retrieve without looking, then check. Spread it across weeks. In this semester level 1 is small — eight objectives — so it should not be taking large amounts of lecture time.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material, not receive it again: sort it, contrast it, put it in their own order, explain it to a classmate. Fifty-five objectives sit here, the largest single group in the semester, and lecture alone will not deliver them.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else.

LEVEL 5 IN 1956, SYNTHESIS

Used four times in this semester, in Modules 4 and 5. It means the student assembles something that was not handed to them — a protocol, a specification, a procedure. Set the criteria in advance and mark against them.

LEVEL 6 — EVALUATION

Give something to judge and criteria to judge it against, then require the criteria to be stated. Small-group critique works well. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching one.

THE DOING LADDER

- **Set** — show the setup, name every item, then have students lay it out themselves.
- **Guided response** — demonstrate once at speed, once slowly, then let them copy while you correct in the moment.
- **Mechanism** — repeated supervised practice with your help withdrawn a little each time. **Most practical hours belong here.**
- **Adaptation** — change one condition and let them work out the adjustment.
- **Origination** — used twice, in Module 3. The student devises a procedure that was never demonstrated.

PART A · CHAPTER 8

Choosing how to check, level by level, and the modular assessment

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half.

- **Level 1** — short questions asking for a list, a name, a definition. Many of them, spread out.
- **Level 2** — questions requiring the material rearranged. If it can be answered word-for-word from notes, it has slipped to level 1.
- **Level 3** — an unseen case. A familiar case measures memory.
- **Level 4** — questions asking for the parts and how they relate.
- **Level 5** — something the student produced, marked against criteria set in advance.
- **Level 6** — give the marks for the reasoning and the criteria, not for the facts repeated inside the answer.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

A viva shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably. With 65 doing objectives in this semester, this matters more here than anywhere. The evidence has to be seen: a checklist ticked while the student works, an OSPE station, an observed performance. A checklist of the steps you will look for, written before the session, is enough.

THE MODULAR ASSESSMENT

Each module carries a modular assessment worth 25 marks per credit — 400 marks across the semester — using two of the module's hours per credit. The curriculum requires a **different method in each module**, with a record of the pattern used.

> **Choose the method against the module's ceiling, not by rotation.** A module whose ceiling is Origination or Synthesis needs an assessment that lets a student produce something. A module that stops at Application does not, and using an elaborate instrument there wastes everyone's time. The modules in Part B print each ceiling for exactly this decision.

PART A · CHAPTER 9

How to use a module and a unit in Part B

Part B has eight modules, each followed by its units — fifty entries in all. Every one begins on its own page.

THE MODULE

1. **The strip** — credits, units, thinking and doing counts, formative marks.
2. **Where the hours go** — the lecture, practical and experiential split for this module, with what each kind of hour can reach and what it must deliver here.
3. **What the module asks students to do** — both ladders, across all its units.
4. **Module learning objectives**, each with the level of its verb. These are what the modular assessment tests.
5. **The units**, with each unit's highest thinking and highest doing level side by side.

THE UNIT

1. **The heading** — objectives, hours, the mix of lecture, practical and experiential, and the course outcomes served.
2. **The syllabus points** as the curriculum lists them.
3. **Both ladders** for this unit.
4. **The objectives grouped by level**, each with its verb, its kind of knowledge, its kind of hour, and what evidence it needs.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Take one level at a time from the grouped objectives and teach upward. Check the method against Chapter 7 and the kind of hour against Chapter 6.

Designing the modular assessment. Start from the module's ceilings and the module learning objectives, then pick an instrument that can actually reach them.

Giving students their bearings. The student panel is written to be read by students. Its first line is the useful one: read the verb, and read the kind of hour.

> **What this book does not do.** It does not rewrite the curriculum and it does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking.

PART B

The modules and units

Fifty entries: eight module overviews, each followed by its units. Every objective appears in the curriculum's own words, sorted into its level so a session can be planned around one level at a time, with the kind of hour printed against each — lecture, practical training or experiential learning. Chapter 9 explains the layout; after that they are meant to be opened at whichever one you are about to teach.

1. **Modules** — hours, ladders, module learning objectives, and the units within
2. **Units** — syllabus points, both ladders, and the objectives grouped by level

↑ Contents

MODULE 1

Pharmaceutics and Pharmaceutical Technology

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	14	10	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Pharmaceutics and Pharmaceutical Technology provides a comprehensive foundation in pharmaceutics and pharmaceutical technology,covering the entire spectrum of pharmaceutical design, formulation, manufacture, and evaluation. It also delves into the pharmaceutical industry, regulatory frameworks, and product development processes. Additionally, the module explores various pharmaceutical

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

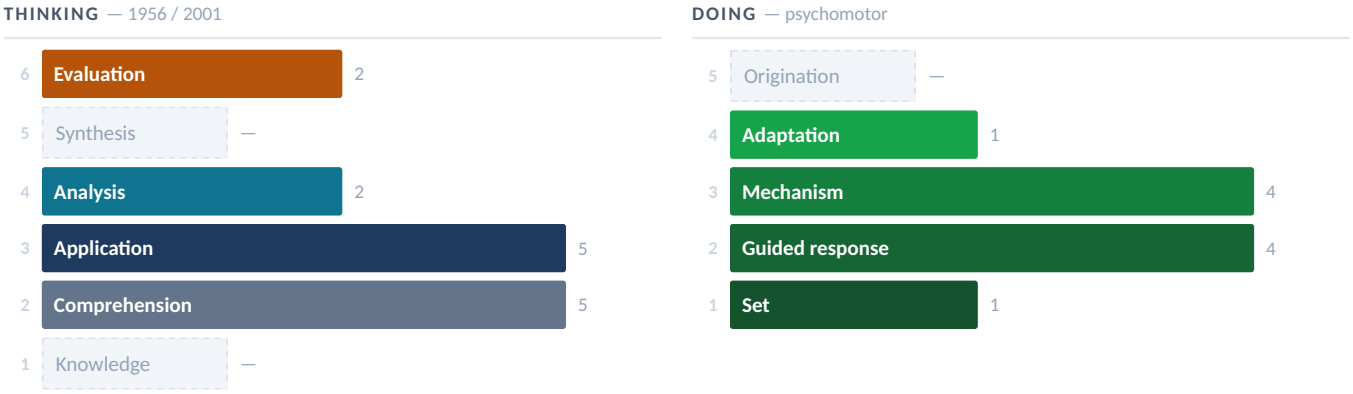
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	8	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Discuss the fundamental principles and scope of pharmaceutics and pharmaceutical technology.	*discuss*	Level 2 Comprehension / Understand

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe the different types of pharmaceutical dosage forms and their formulation principles.	*describe*	Level 2 Comprehension / Understand
3	Discuss the importance of Good Manufacturing Practices (GMP) and quality control in pharmaceutical manufacturing.	*discuss*	Level 2 Comprehension / Understand
4	Apply biopharmaceutic principles to pharmaceutical product development and optimization.	*apply*	Level 3 Application / Apply
5	Illustrate the pharmaceutical product development process and regulatory frameworks governing the pharmaceutical industry.	*illustrate*	Level 2 Comprehension / Understand

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Pharmaceutics	6	12 h	Evaluation	Adaptation
2	Pharmaceutical Dosage Forms	4	8 h	Analysis	Mechanism
3	Pharmaceutical Formulation	5	16 h	Application	Mechanism
4	Pharmaceutical Manufacturing and Quality Control	5	10 h	Evaluation	Mechanism
5	Biopharmaceutics	4	10 h	Application	Mechanism

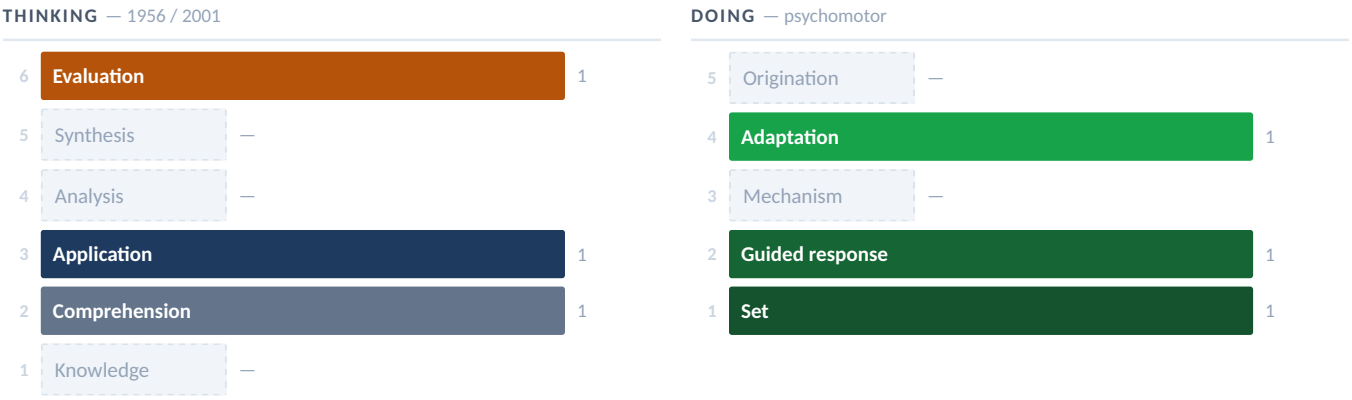
MODULE 1 · UNIT 1

Introduction to Pharmaceutics

6 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 2 | Outcomes: CO 1, CO 2, CO 4, CO 5, CO 7

- Definition and scope of pharmaceutics
- Historical development of pharmaceutics
- Pharmaceutical industry overview
- Regulatory frameworks (FDA, etc.)
- Pharmaceutical product development process

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Define pharmaceutics and describe its scope and historical development.	*define*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *define* is what will be asked.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Enlist and evaluate challenges and solutions in the pharmaceutical product development process.	*enlist*	Procedural °	Experiential 3h	Written or spoken. Pitch at **level 1** — the verb *enlist* is what will be asked.
DOING — LEVEL 1 · SET — 1				
Demonstrate regulatory frameworks (FDA, etc.).	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate pharmaceutical product development process.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 1				
Identify applications suiting various regulatory frameworks.	*identify*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

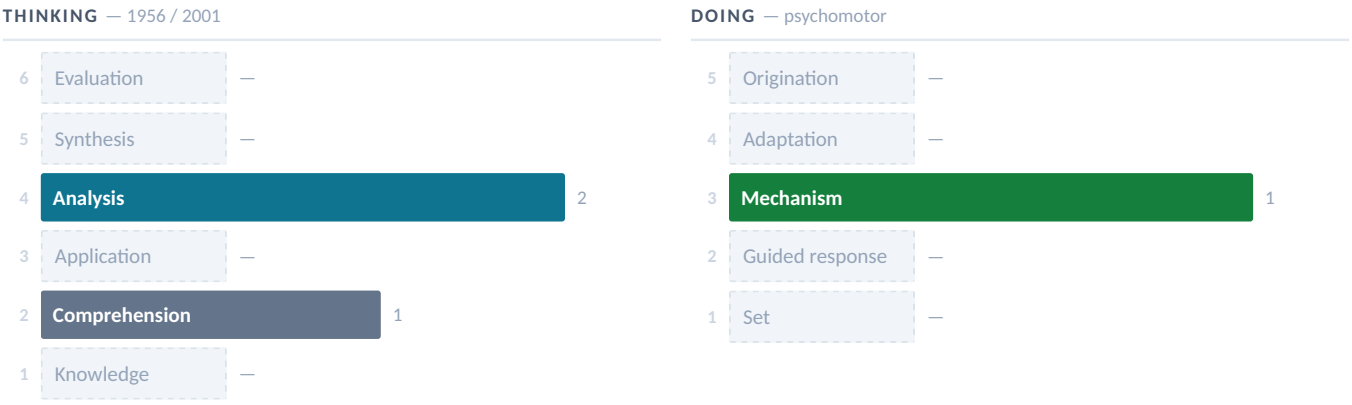
MODULE 1 · UNIT 2

Pharmaceutical Dosage Forms

4 objectives | 8 hours | Practical 2 · Lecture 1 · Experiential 1 | Outcomes: CO 1, CO 5

- Classification of dosage forms
- Solid dosage forms (tablets, capsules, etc.)
- Liquid dosage forms (solutions, suspensions, etc.)
- Semi-solid dosage forms (ointments, creams, etc.)
- Parenteral dosage forms (injections, infusions, etc.)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Classify various pharmaceutical dosage forms.	*classify*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Identify and categorise classical Ayurvedic dosage forms.	*identify*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
Identify and categorise modified Ayurvedic dosage forms.	*identify*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Explore, identify, and collect various classical and modified pharmaceutical dosage forms.	*explore*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 1 · UNIT 3

Pharmaceutical Formulation

5 objectives | 16 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 1, CO 5

- Principles of pharmaceutical formulation
- Excipients and their functions
- Solid dosage forms
- Liquid and semi-solid dosage forms
- Parenteral dosage forms

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss formulation of parenteral dosage forms.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the principles of pharmaceutical formulations, excipients, and their functions.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Formulate Solid Dosage Forms selecting appropriate excipients.	*formulate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Formulate liquid Dosage Forms selecting appropriate excipients.	*formulate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Identify an Appropriate Excipient According to Solid and Liquid Dosage Form.	*identify*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

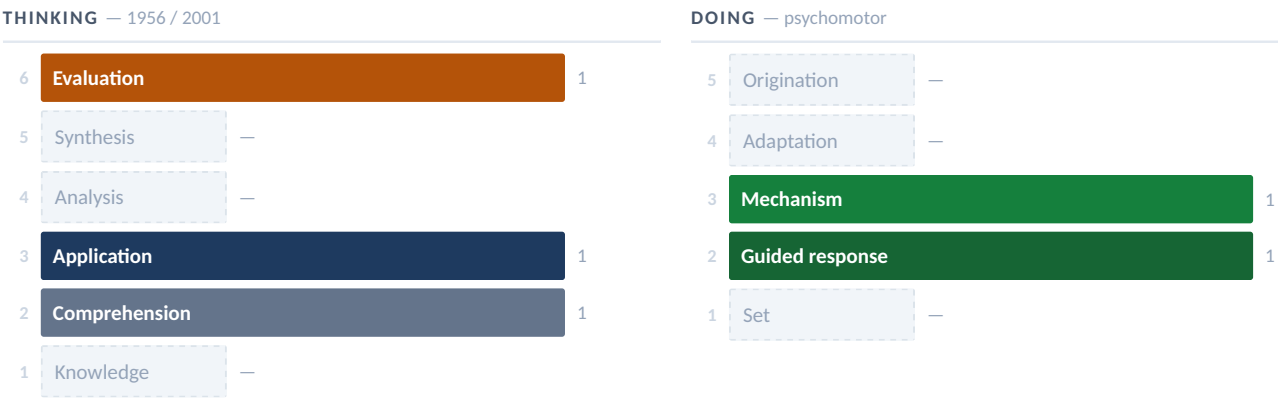
MODULE 1 · UNIT 4

Pharmaceutical Manufacturing and Quality Control

5 objectives | 10 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 3, CO 4

Good Manufacturing Practices (GMP& cGMP)
Quality control and quality assurance
Packaging and labelling
Portability and storage

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Good Manufacturing Practices (GMP& cGMP).	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss pharmaceutical stability and shelf life.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Evaluate GMP audit and compliance.	*evaluate*	Conceptual °	Practical 2h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate, design and evaluate pharmaceutical packing and labelling.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Design and develop Pharmaceutical Packaging systems and labelling.	*design*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 1 · UNIT 5

Biopharmaceutics

4 objectives | 10 hours | Lecture 3 · Experiential 1 | Outcomes: CO 4, CO 5, CO 6, CO 7, CO 8

- Biopharmaceutics and pharmacokinetics
- Bioavailability and bioequivalence
- Pharmaceutical dissolution and absorption
- Pharmaceutical product development and optimization

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe biopharmaceutics and pharmacokinetics.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Describe bioavailability and bioequivalence.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
Discuss pharmaceutical product development and optimization.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Design and Optimize Pharmaceutical Formulation for Enhanced Bioavailability.	*design*	Procedural	Experiential 7h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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MODULE 2

Databases and Information Systems for Metal, Mineral, and Plant Research

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	17	4	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Databases and Information Systems for Metal, Mineral, and Plant Research, provides a comprehensive introduction to databases and information systems, focusing on metal, mineral, and plant research. This module will provide the fundamental principles of databases, data retrieval languages, and various databases relevant to the field, including metal, mineral, and plant databas

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

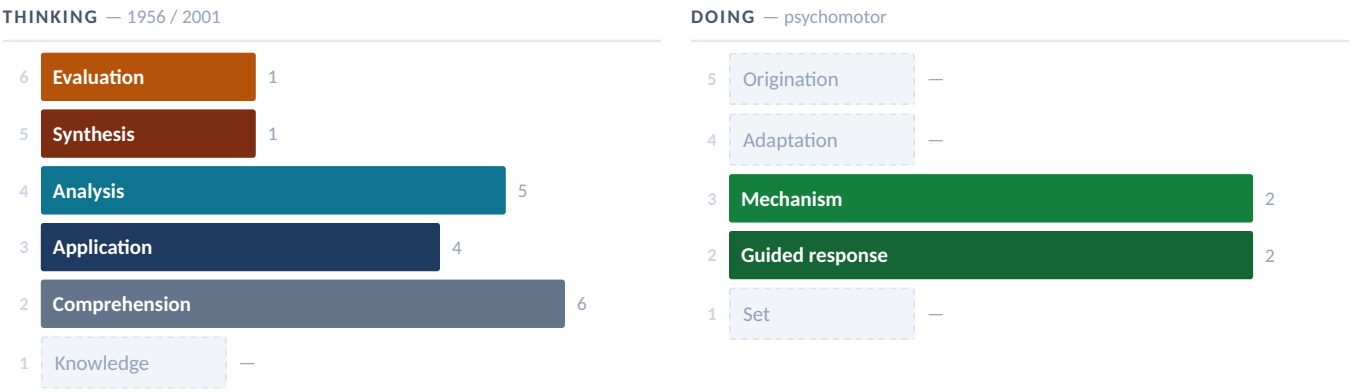
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	7	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	5	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the fundamental concepts of databases and information systems, including types of databases and database management systems.	*describe*	Level 2 Comprehension / Understand
2	Acess metal, mineral, and plant databases for research and development, including advanced searching techniques and data analysis.	*acess*	Level ? —
3	Integrate multiple databases and apply database applications in research, industry, and decision-making.	*integrate*	Level 5 Synthesis / Create
4	Evaluate the challenges and limitations of databases in metal, mineral, and plant research.	*evaluate*	Level 6 Evaluation / Evaluate
5	Design and develop a database application for metal, mineral, or plant research, incorporating emerging trends and technologies.	*design*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Databases and Information Systems	5	12 h	Application	Mechanism
2	Metal and Mineral Databases	6	16 h	Synthesis	—
3	Plant Databases	4	12 h	Analysis	—
4	Database application	4	12 h	Evaluation	Mechanism
5	Challenges and limitation of Databases	3	4 h	Analysis	—

MODULE 2 · UNIT 1

Introduction to Databases and Information Systems

5 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 1

- Overview of databases and information systems
- Types of databases (relational, object-oriented, etc.)
- Database management systems (DBMS)
- Data retrieval and querying languages (SQL, etc.)
- Introduction to metal, mineral, and plant databases

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS).	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Demonstrate various searching techniques of databases.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
Demonstrate Modelling, Querying, and Analytics of Data.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Design and Implement a Simple Database for a Plant Nursery.	*design*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

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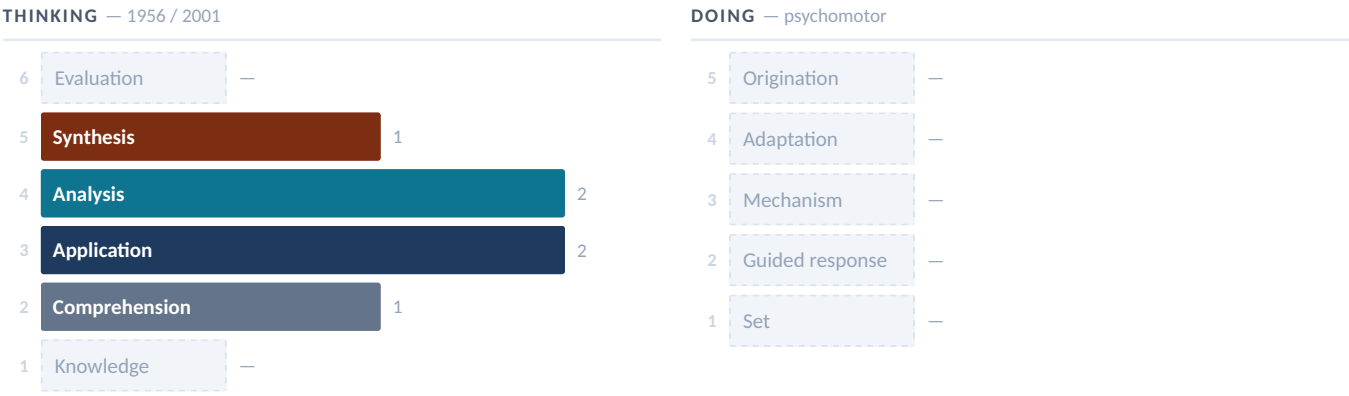
MODULE 2 · UNIT 2

Metal and Mineral Databases

6 objectives | 16 hours | Practical 3 · Lecture 2 · Experiential 1 | Outcomes: CO 3, CO 4

- Overview of metal and mineral databases
- USGS Mineral Database
- Mineral Database
- Webmineral Mineral Database
- Introduction to Metal Databases

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe overview of metal and mineral databases, USGS (U.S. Geological Survey).	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Discuss Mineral and metal Databases, including their applications and significance.	*discuss*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
Search, explore and analyze Metal Database and Applications.	*search*	Conceptual °	Practical 3h	Written or spoken. Pitch at **level 4** — the verb *search* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Analyze USGS Mineral Database.	*analyze*	Conceptual °	Practical 2h	Written or spoken answer, pitched at the coded level.
Analyse Web mineral Database and Metal Database.	*analyse*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Analysis and integration of Web mineral Database.	*analyse*	Procedural °	Practical 3h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

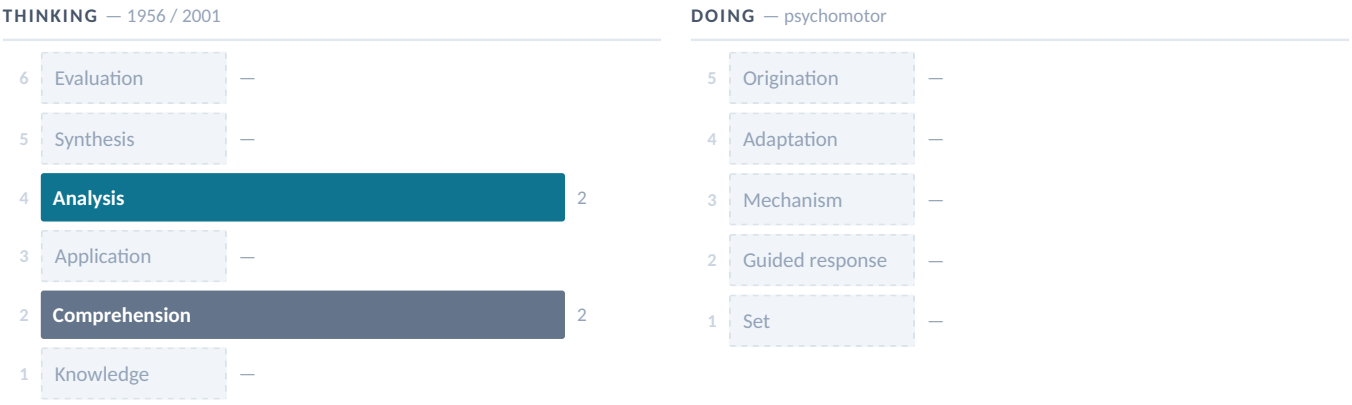
MODULE 2 · UNIT 3

Plant Databases

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 3, CO 4

- Overview of plant databases
- The Plant List (TPL)
- International Plant Names Index (IPNI)
- Royal Botanic Gardens, Kew (RBG Kew) databases
- Introduction to plant genetic databases (e.g., PlantGDB)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Illustrate overview of plant databases, The Plant List (TPL), and International Plant Names Index (IPNI).	*illustrate*	Factual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe Royal Botanic Gardens, Kew (RBG Kew) databases and introduction to plant genetic databases (e.g., PlantGDB).	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Demonstrate techniques to explore Plant Databases.	*demonstrate*	Procedural	Practical 4h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
Explore and analyse plant database.	*explore*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
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MODULE 2 · UNIT 4

Database application

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 3, CO 4

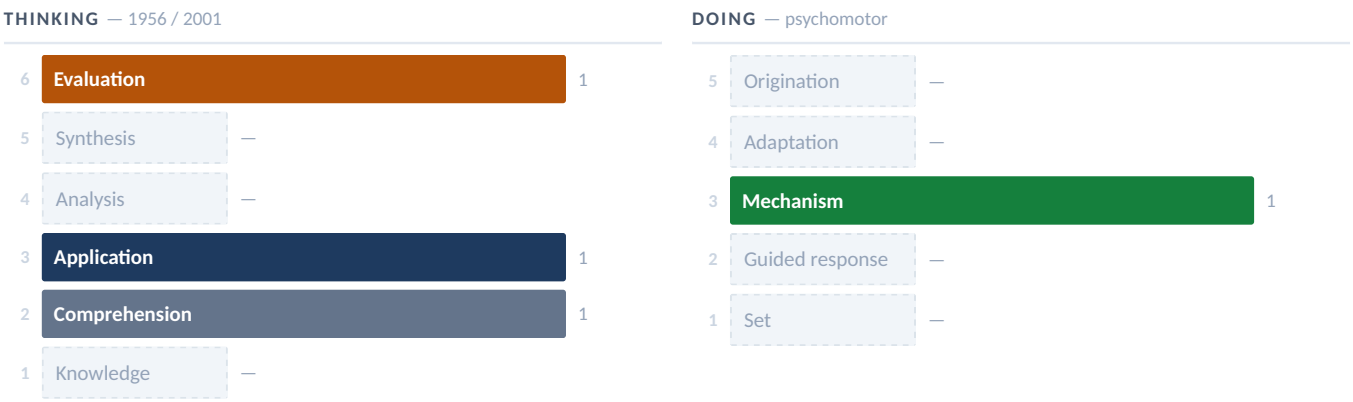
Database applications in research and industry

Using databases for decision-making and policy development

Future directions: emerging trends and technologies in database development

Group project: developing a database application for metal, mineral, or plant research

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss database applications in research and industry, and use databases for decision-making and policy development.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Analyze future directions in database development and applications, including emerging trends and technologies, and group project guidelines.	*analyze*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 4** — the verb *analyze* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.	*analyse*	Conceptual °	Practical 4h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research.	*apply*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

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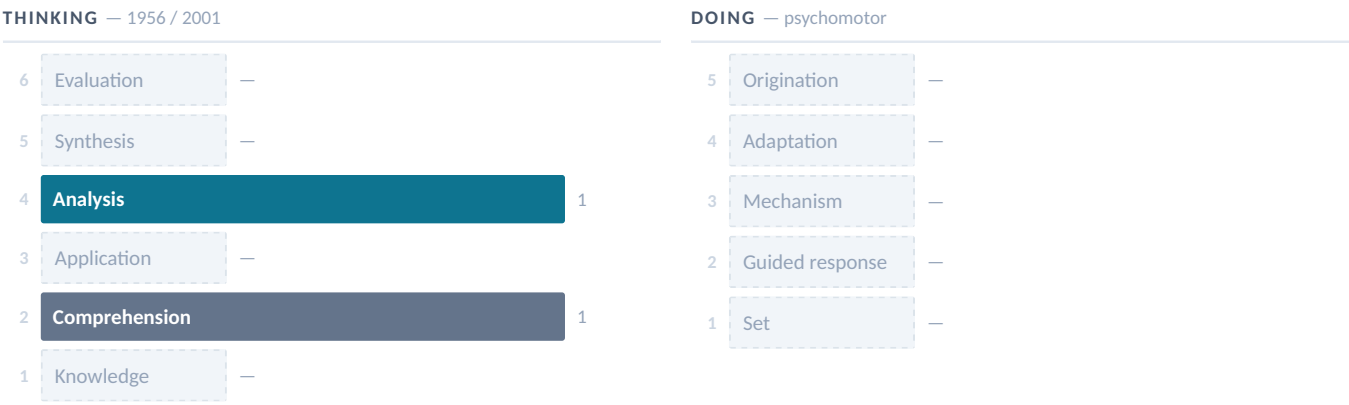
MODULE 2 · UNIT 5

Challenges and limitation of Databases

3 objectives | 4 hours | Lecture 2 · Experiential 1 | Outcomes: CO 3, CO 4

Challenges in Pharmaceutical Formulation Development
Limitations of In Vitro and In Vivo Testing
Regulatory Challenges in Pharmaceutical Product Approval
Scalability and Manufacturing Challenges

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Identify Database-Related Challenges in Pharmaceutical Development.	*identify*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Discuss data Quality, Integration, Security, Scalability, and Regulatory Compliance.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1				
Perform a role play related to the challenges and limitations of databases.	*perform*	Conceptual	Experiential 2h	**Watch over time.** Conduct across the module, not a single answer.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3

Essential Chemistry :- Unlocking the Rasashastra Techniques

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	12	9	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Essential Chemistry - Unlocking the Rasashastra Techniques integrates fundamental chemistry principles with traditional Ayurvedic practices,enhancing understanding and application of Rasashastra techniques. By exploring the connections between chemistry and Ayurveda, learners will gain adeeper 2 10 20 30 60 understanding of the processing methods and transformations involved in Rasasha

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

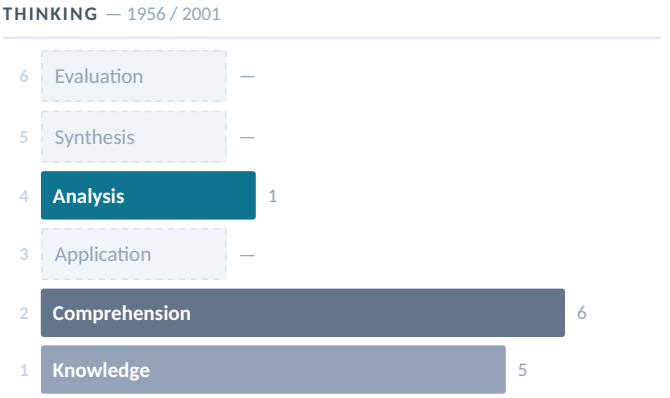
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

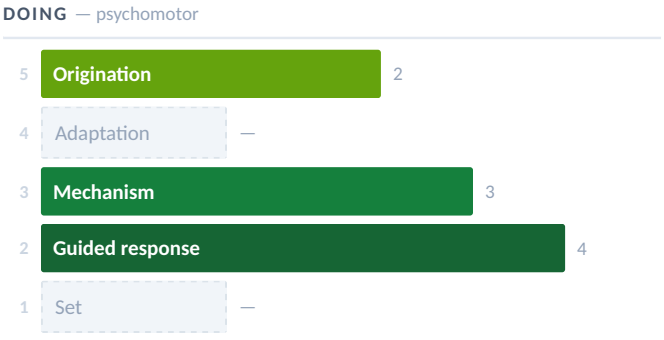
KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.





THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Atomic, Molecular Structure, and Periodic Table Elements	3	12 h	Knowledge	Mechanism
2	Specialized Chemistry Fields-An Outline	2	6 h	Comprehension	—
3	Organic Chemistry	3	8 h	Analysis	—
4	Phytochemistry	7	16 h	Comprehension	Mechanism
5	Metallurgy	6	14 h	Comprehension	Origination

MODULE 3 · UNIT 1

Atomic, Molecular Structure, and Periodic Table Elements

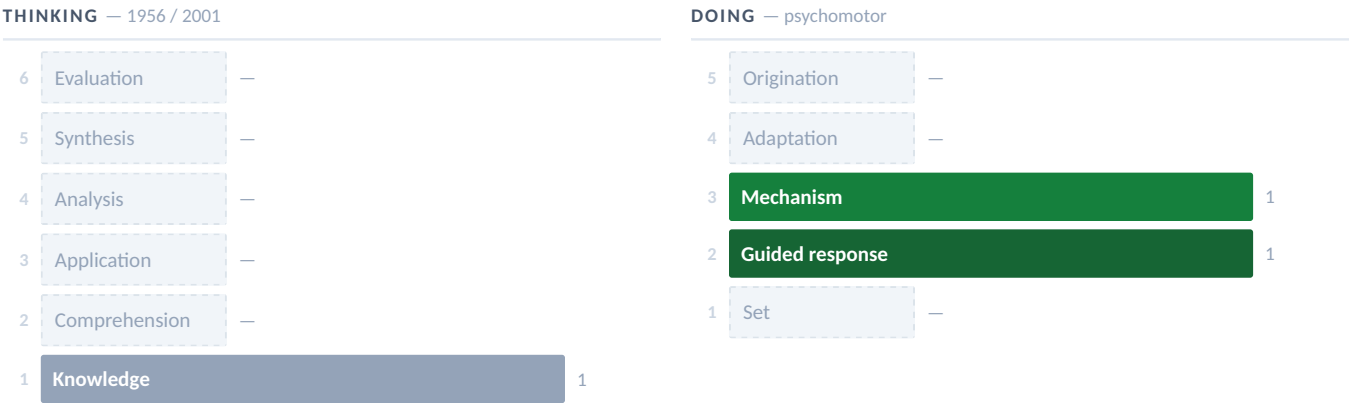
3 objectives | 12 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 8

Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha nirmana.

Atomic structure, Periodic table, Chemical bonding, Isomerism - Application of these principles in Shodhana, Marana, Amruteekarana and Chaturvidha Rasayana

Periodic Table, Main group elements, Transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability,

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the role of basic chemistry principles and their application in Rasashastra.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana.	*analyse*	Conceptual	Practical 5h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals.	*apply*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 2

Specialized Chemistry Fields-An Outline

2 objectives | 6 hours | Lecture 1 · Practical 1 | Outcomes: CO 8

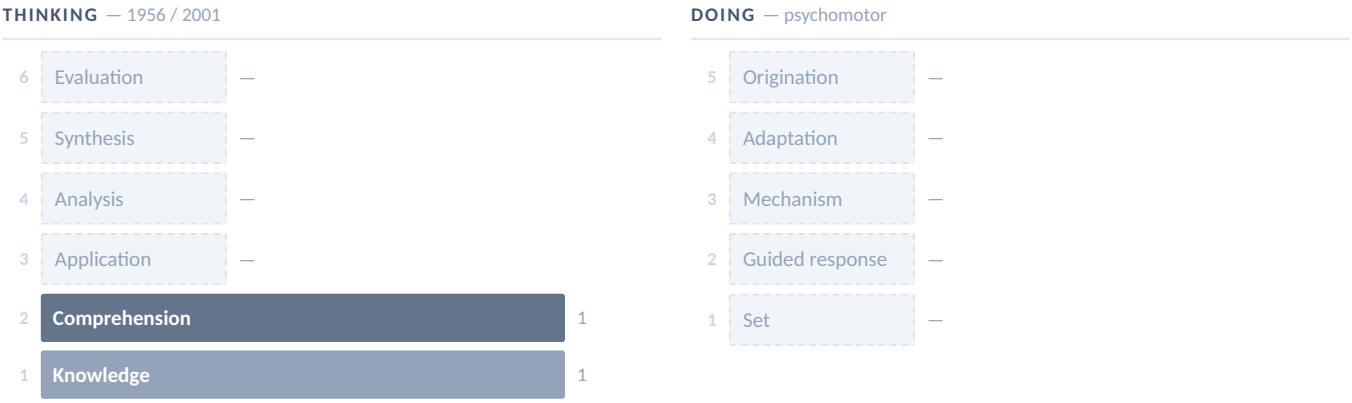
Crystallography- detoxification, transformation of compound

Organometallic chemistry- Chaturvidha Rasayana

Bioinorganic chemistry- Amruteekarana (Metal-Carbon bond-stable compound formation)

Application in Shodhana, Marana, Amruteekarana, Chaturvidha Rasayana

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures.	*demonstrate*	Conceptual + Procedural °	Practical 5h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 3

Organic Chemistry

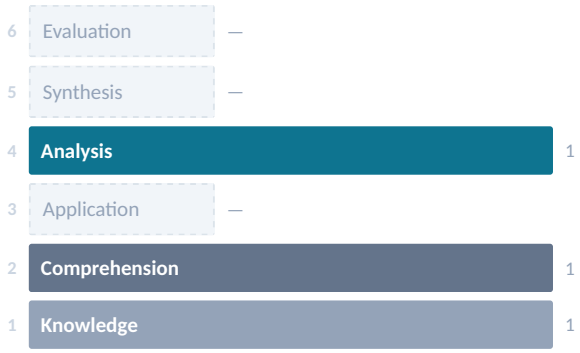
3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 8

Structure of organic molecules: understanding of Herbal components, Interaction of herbs and animal products with metals for detoxification, compound formation, and compound stability.

Functional groups: Reactivity and properties of organic molecules influencing compound action

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the structure of organic molecules.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the interaction of herbs and animal products with metals.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy.	*explore*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 4

Phytochemistry

7 objectives | 16 hours | Lecture 3 · Practical 2 · Experiential 2 | Outcomes: CO 8

- Definition and significance of phytochemistry
- Overview of plant-based bioactive compounds
- Classes of Phytochemicals
- Extraction methods
- Phytochemical screening

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3				
Describe the significance of phytochemistry.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe the overview of plant-based bioactive components.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss different types of extraction methods including solvent selection.	*discuss*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Demonstrate different types of extraction methods including solvent selection.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 2				
Demonstrate extraction of different herbal formulations using various extraction techniques and solvents.	*demonstrate*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.
Identify phytoconstituents in Herbs and Herbal Formulations: A.	*identify*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

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MODULE 3 · UNIT 5

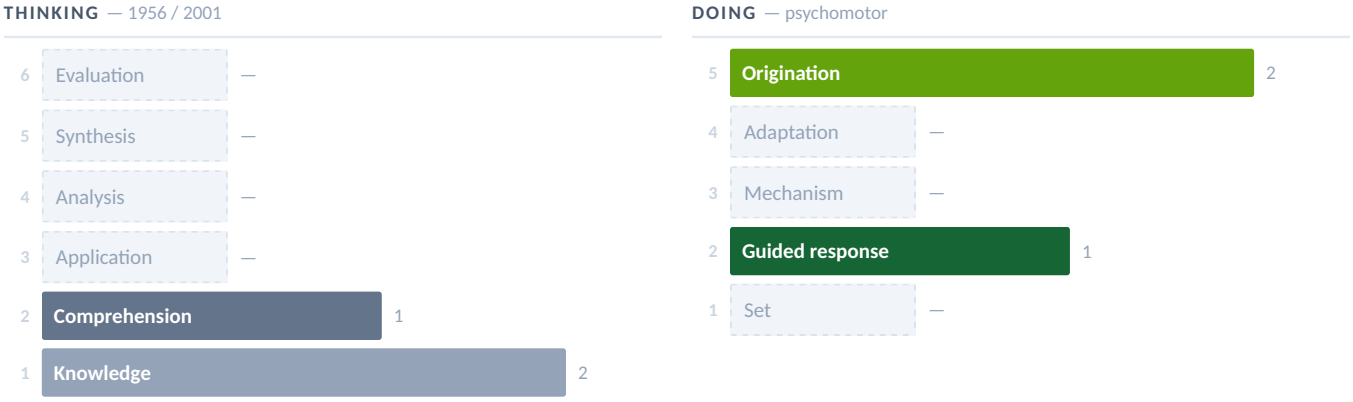
Metallurgy

6 objectives | 14 hours | Lecture 3 · Experiential 2 · Practical 1 | Outcomes: CO 8

Introduction to metallurgy and general physicochemical methods of extraction of metals from ores

Common ores used in Rasashastra and their Satvapatana from metallurgical point of view

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 2				
Define metallurgy and describe its scope.	*define*	Factual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe general physicochemical methods of extraction of metals from ores.	*describe*	Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss common ores used in Rasashastra and their Satvapatana with metallurgical point of view.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate different types of metals and their properties.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 5 · ORIGINATION — 2				
Differentiate various types of metals based on their properties.	*differentiate*	Conceptual + Procedural °	Experiential 4h	**Watch them do it.** Alter a condition and watch what they change.
Differentiate metals using basic metal identification techniques.	*differentiate*	Procedural	Experiential 4h	**Watch them do it.** Alter a condition and watch what they change.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4

Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	4	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Concept, understanding, application, SOPs, and chemical changes of Shodhana and Marana procedures provides a comprehensive understanding of Shodhana (purification/ Detoxification/ modification/Potentiating) and Maran (incineration) procedures in Ayurvedic pharmaceutical manufacturing. It covers the fundamental principles, applications, and standard opera

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

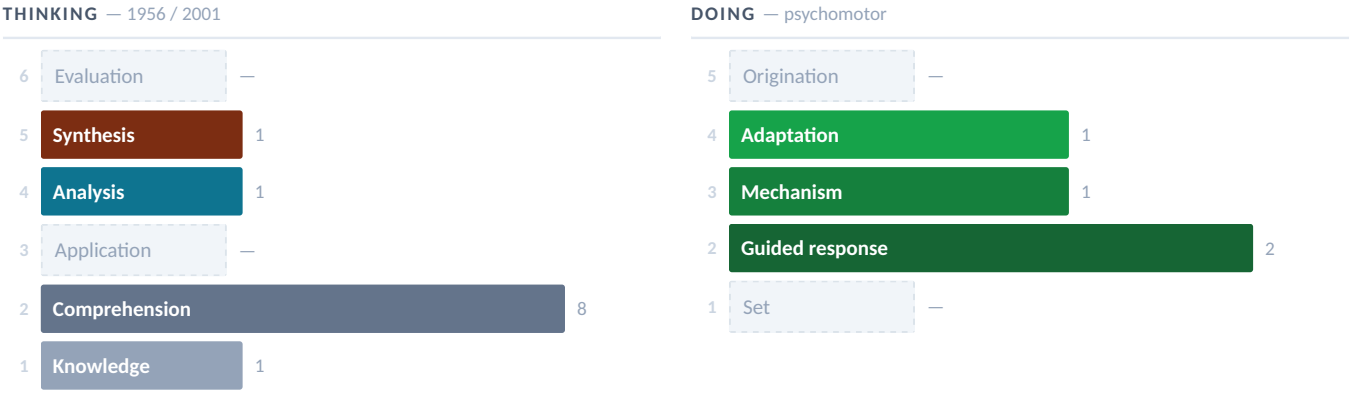
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	2	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	3	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Demonstrate a comprehensive understanding of Shodhana (purification/ Detoxification/ modification/Potentiating) and Maran (incineration) procedures.	*demonstrate*	Level 3 Application / Apply

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe the fundamental principles, applications, and standard operating procedures (SOPs) of Shodhana and Marana processes.	*describe*	Level 2 Comprehension / Understand
3	Describe the chemical changes involved in the Shodhana and Marana processes.	*describe*	Level 2 Comprehension / Understand
4	Analyze real-world case studies of Shodhana and Marana procedures.	*analyze*	Level 4 Analysis / Analyse
5	Apply troubleshooting techniques for common issues in Shodhana and Marana processes.	*apply*	Level 3 Application / Apply
6	Plan future directions in Shodhana and Marana procedures.	*plan*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Shodhana and Marana Procedures	3	3 h	Analysis	—
2	SOP of Shodhana Processes	4	22 h	Comprehension	Adaptation
3	SOP of Marana process	4	22 h	Comprehension	Mechanism
4	Chemical Changes involved in Shodhana and Marana processes	1	1 h	Comprehension	—
5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	8 h	Synthesis	—

MODULE 4 · UNIT 1

Shodhana and Marana Procedures

3 objectives | 3 hours | Lecture 3 | Outcomes: CO 1, CO 2, CO 7

Overview of Shodhana and Marana procedures

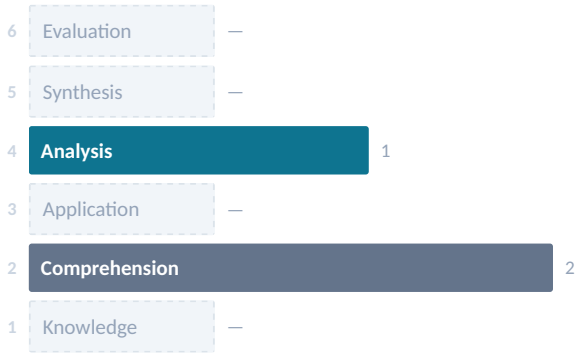
Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing

Principles of Shodhana and Marana procedures

Applications of Shodhan and Maran procedures in Ayurvedic pharmaceutical manufacturing

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing.	*analyse*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

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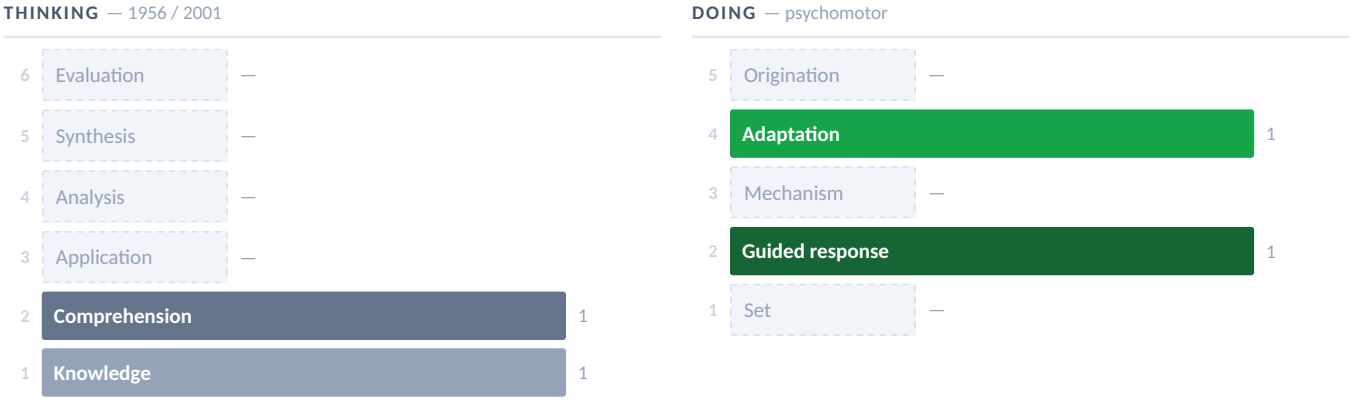
MODULE 4 · UNIT 2

SOP of Shodhana Processes

4 objectives | 22 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 7, CO 8

Types of Shodhana procedures (physical, chemical, biological)
Methods of Shodhana (filtration, sedimentation, distillation, etc.)
SOPs for Shodhana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe types of Shodhana procedures (physical, chemical, biological).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss methods of Shodhana (filtration, sedimentation, distillation, etc.) and its SOPs.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Shodhana procedure of Rasadravyas using various methods.	*demonstrate*	Procedural	Practical 10h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 1				
Perform Shodhana of Rasadravyas with various methods.	*perform*	Procedural	Experiential 10h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 3

SOP of Marana process

4 objectives | 22 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 7

Types of Marana procedures (incineration, calcination, etc.)
Methods of Marana (oxidation, thermal decomposition, etc.)
SOPs for Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe types of Marana procedures (incineration, calcination, etc.).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss methods of Marana (oxidation, thermal decomposition, etc.) and its SOPs.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Marana procedure of Rasadravyas with different methods.	*demonstrate*	Procedural	Practical 10h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform Marana procedure of Rasadravyas with different methods.	*perform*	Procedural	Experiential 10h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 4

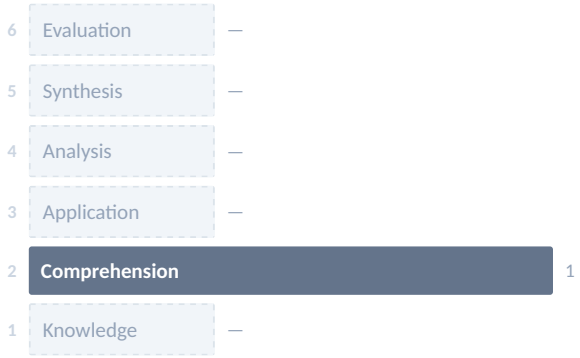
Chemical Changes involved in Shodhana and Marana processes

1 objectives | 1 hours | Lecture 1 | Outcomes: CO 1, CO 7

Chemical changes involved in Shodhan procedures	Chemical Changes involved in Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss chemical changes involved in Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 5

Shodhana and Marana- Case Studies, troubleshooting and emerging trends

3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 1, CO 7, CO 8

Case studies of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing

Troubleshooting and optimization of Shodhana and Marana procedures

Future directions and emerging trends in Shodhana and Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe Troubleshooting and optimization of Shodhana and Marana procedures.	*describe*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss future directions and emerging trends in Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies.	*analyse*	Procedural °	Experiential 6h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5

Principles, specification, SOPs of handling of classical and advanced instruments

CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	7	20	9	75
90 notional hours	29 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Principles, specifications, SOPs for handling classical and advanced instruments describes the principles, specifications, and SOPs for handling classical instruments like Puta and Musha, and their modern aspects. It also deals with the principles, types, and applications of chromatography, spectroscopy, and microscopy in the standardization of Ayurvedic formulations, demon

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

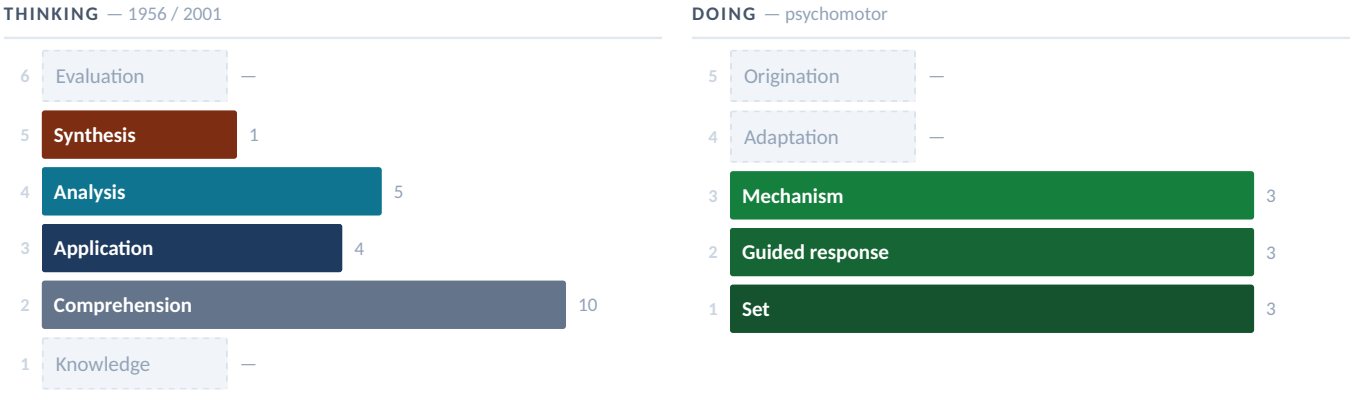
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	15 h	15	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	30 h	7	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	45 h	7	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 6 hours of the 45 shown, worth 75 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the principles, specifications, and SOPs for handling classical instruments like Puta and Musha, and their modern aspects.	*describe*	Level 2 Comprehension / Understand
2	Demonstrate the use of advanced characterization instruments, such as PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM analysis.	*demonstrate*	Level 3 Application / Apply
3	Develop SOPs for calibration, troubleshooting, and maintenance of various instruments.	*develop*	Level 5 Synthesis / Create
4	Integrate the knowledge of instrumental analysis for the standardization and quality control of Ayurvedic medicines	*integrate*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Puta and its modern aspects	4	14 h	Synthesis	Guided response
2	Musha and its modern aspects	4	14 h	Analysis	Mechanism
3	Chromatography	5	17 h	Application	Mechanism
4	Spectroscopy	4	9 h	Analysis	Set
5	Microscopy	4	12 h	Comprehension	Mechanism
6	Calibration techniques	3	7 h	Analysis	Set
7	Overview of characterization instruments	5	11 h	Analysis	—

MODULE 5 · UNIT 1

Puta and its modern aspects

4 objectives | 14 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 7

Introduction, principles, and specifications of Puta
SOPs for handling Puta
Modern aspects of Puta (automation, computerization, etc.)
Troubleshooting and maintenance of Puta

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the principles and specifications of Puta.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse modern aspects of Puta.	*analyse*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Develop Standard Operative Procedures (SOPs)for given Puta.	*develop*	Procedural	Experiential 8h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate selected Puta to give the knowledge of Standard Operative Procedure.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

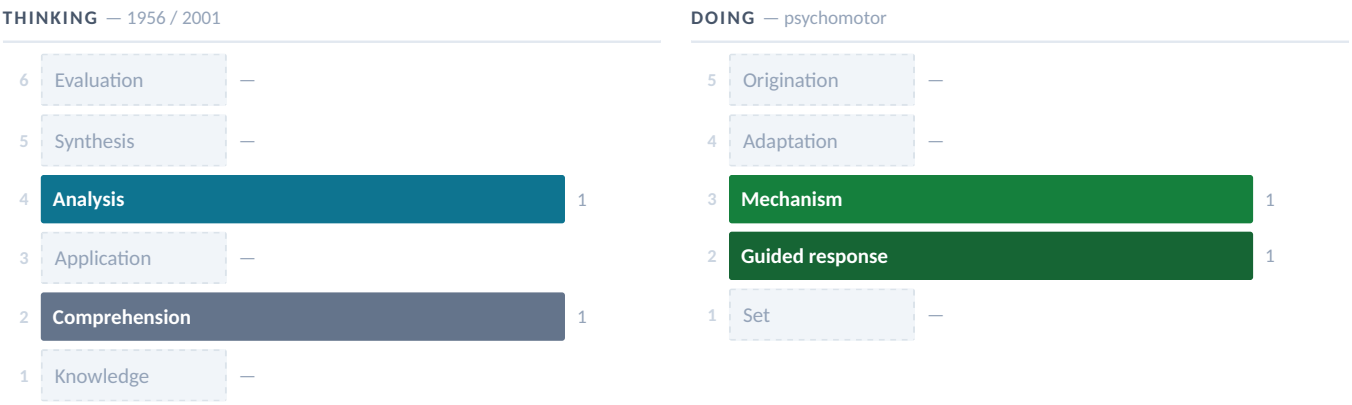
MODULE 5 · UNIT 2

Musha and its modern aspects

4 objectives | 14 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 7, CO 8

Introduction, principles, and specifications of Puta.
SOPs for handling Musha
Modern aspects of Musha (automation, computerization, etc.)
Troubleshooting and maintenance of Musha

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the principles and specifications of Musha.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse modern aspects of Musha with special reference to crucibles.	*analyse*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate the handling of Musha following the Standard Operating Procedure (SOP).	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Develop Standard Operative Procedure for handling Musha.	*develop*	Procedural	Experiential 8h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 3

Chromatography

5 objectives | 17 hours | Lecture 3 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Scope and applications of chromatography in herbal drug standardization (identification of active compounds, adulterant detection, phytochemical analysis, and quality as

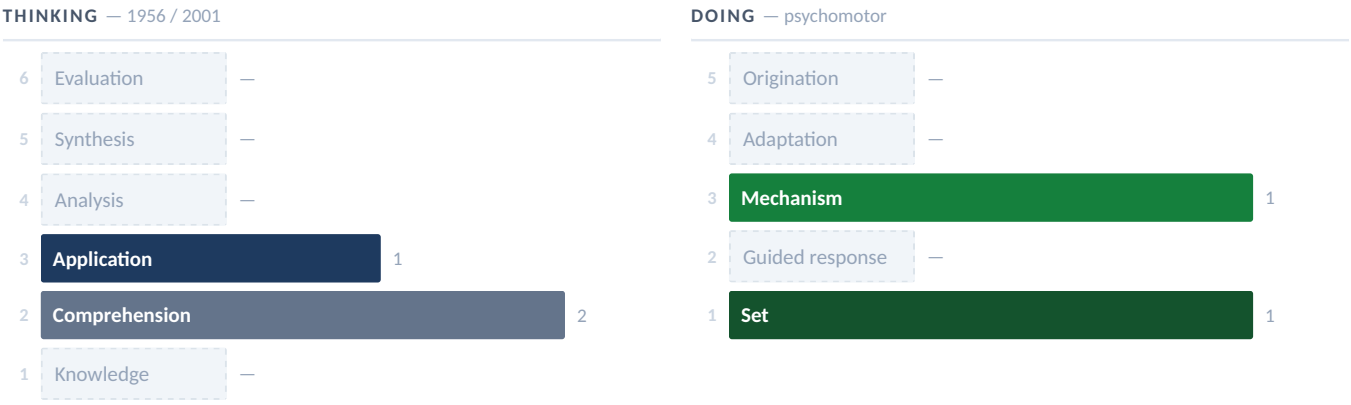
Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS, and GC-MS.

Instrumentation of chromatography, including equipment and tools used in different types of chromatography.

Data analysis in chromatography, including interpretation of chromatograms and spectra.

Chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the instrumentation of chromatography, including equipment and tools used in different types of chromatography.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss the scope and applications of chromatography in herbal drug standardization.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 1 · SET — 1				
Demonstrate chromatography techniques to separate, and identify components and interpretation of results.	*demonstrate*	Procedural	Practical 6h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform chromatography to separate and identify components and interpret the results.	*perform*	Procedural	Experiential 8h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 4

Spectroscopy

4 objectives | 9 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 8

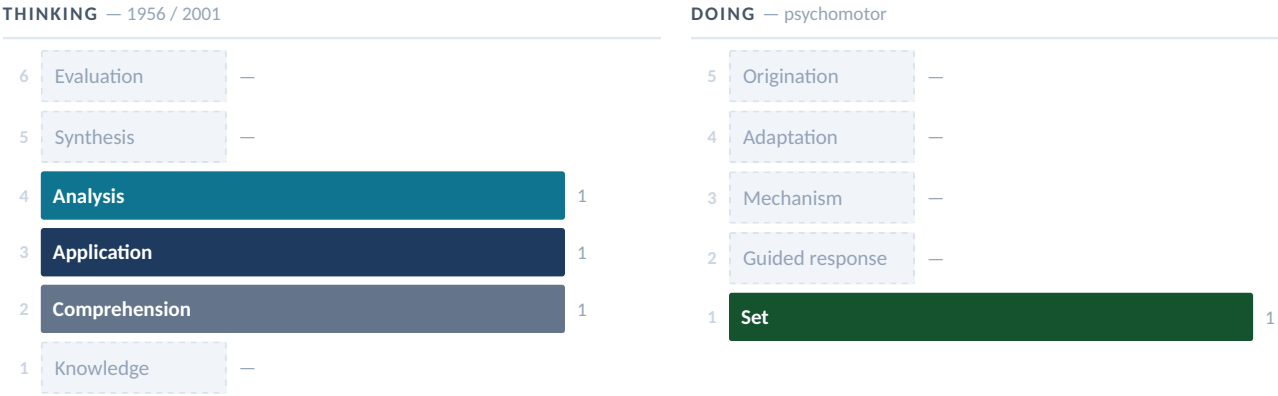
Introduction, principles, and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy.

Specifications of spectroscopy instruments

SOPs for handling spectroscopy instruments

Troubleshooting and maintenance of spectroscopy instruments

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss spectroscopy's principles and types (IR, UV, NMR, AAS, ICP, etc.).	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the specifications of spectroscopy instruments.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Explore and analyze published articles on spectroscopic instruments.	*explore*	Conceptual °	Experiential 4h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 1 · SET — 1				
Demonstration of spectroscopic instruments through Audiovisual aids.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 5

Microscopy

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 8

Introduction, principles, and types of microscopy (optical, petrological, electron, etc.)

Specifications of microscopy instruments

SOPs for handling microscopy instruments

Use of micrometer in microscopy

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the principles, types, and specifications of microscopy instruments.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe the Specifications of microscopy instruments.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate handling of petrological and trinocular microscopes.	*demonstrate*	Procedural	Practical 6h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform hands-on practice on the trinocular microscope.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 6

Calibration techniques

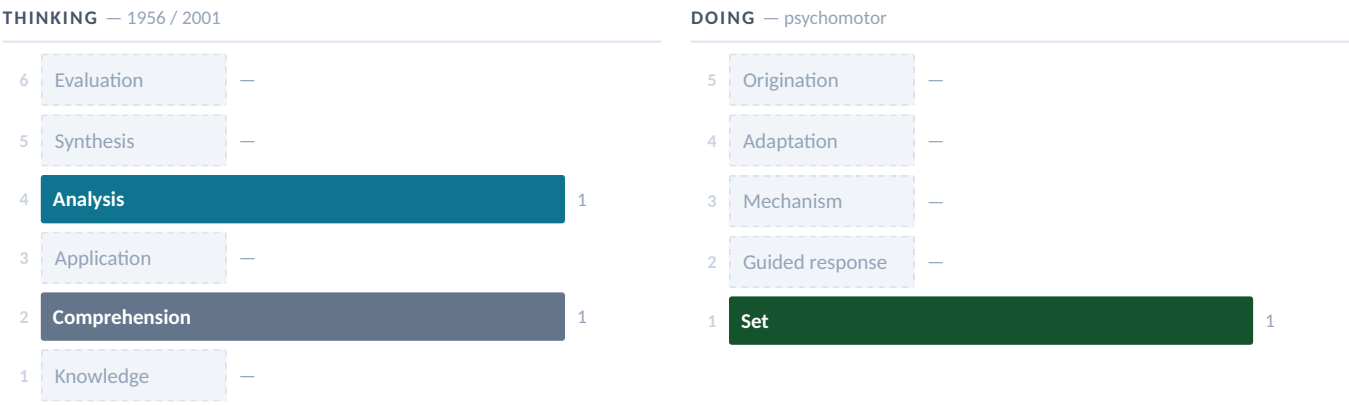
3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Introduction, principles, and types of calibration (instrument calibration, method calibration, etc.)

SOPs for calibration

Troubleshooting and maintenance of calibration instruments

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss principles and types of calibration.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse articles related to challenges in the calibration of equipment.	*analyse*	Conceptual °	Experiential 3h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 1 · SET — 1				
Demonstrate calibration of various instruments present in the department.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 7

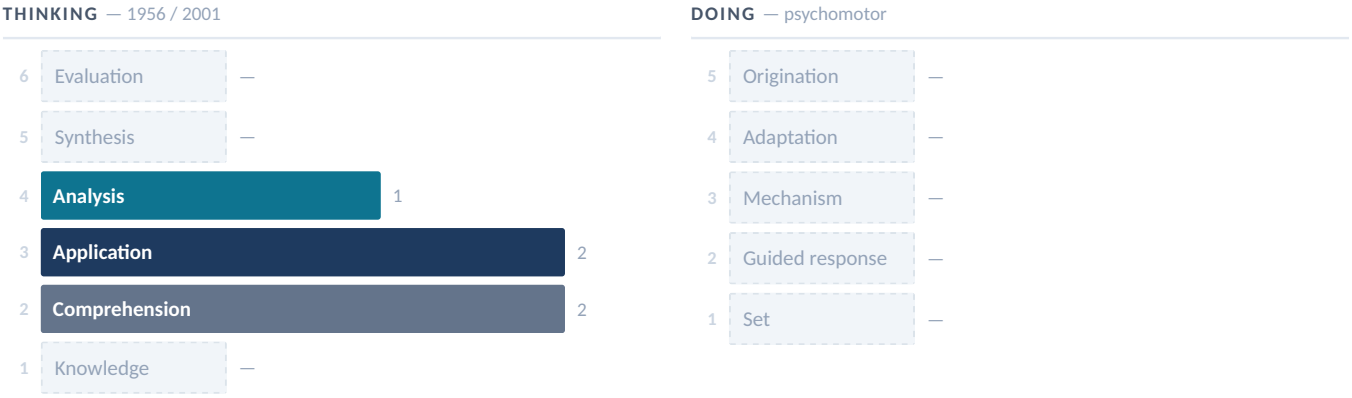
Overview of characterization instruments

5 objectives | 11 hours | Lecture 3 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Overview of advanced characterization instruments and their significance in the standardization of Ayurvedic medicines.

Principles and applications of working mechanism and Applications of instrumental methods: PSA, XRD, XRF, SEM-E-Dax, TEM, TGA, DTA, VSM analysis

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss overview of Advanced Characterization Instruments and their significance in the standardization of Ayurvedic medicines.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss principles and Applications of Spectroscopic and X-ray techniques (PSA, XRD and XRF techniques).	*discuss*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Describe the Principles and Applications of Microscopic and Thermal Techniques(SEM-EDAX, TEM, TGA, DTA, and VSM).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
Demonstrate characterization instruments with Audiovisual aids.	*demonstrate*	Procedural °	Practical 4h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse published articles on advanced instruments used for characterization.	*analyse*	Conceptual °	Experiential 4h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6

Animal house and Animal handling technique

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	5	7	6	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

Tehniques, emphasizing animal care and welfare. It covers the principles of animal house design, sanitation, and hygiene practices, as well as species-specific handling techniques, animal health, and disease prevention. The module also explores advanced animal handling techniques, animal welfare, and research ethics. 1 5 10 15 30

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	5 h	5	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	10 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	15 h	3	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 2 hours of the 15 shown, worth 25 marks.

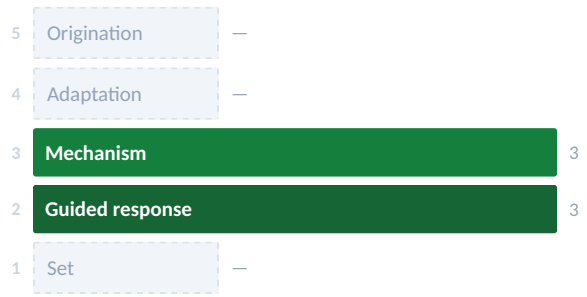
WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the principles of animal care and welfare, and apply them to animal house management.	*describe*	Level 2 Comprehension / Understand

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Demonstrate handling techniques for various animal species, including small and large animals.	*demonstrate*	Level 3 Application / Apply
3	Identify and prevent common diseases in laboratory animals, and implement disease control measures.	*identify*	Level 1 Knowledge / Remember
4	Apply advanced animal handling techniques and promote animal welfare and ethics in research.	*apply*	Level 3 Application / Apply
5	Develop skills in animal behavior observation, enrichment techniques, and humane endpoints and euthanasia methods.	*develop*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to animal house management	3	7 h	Comprehension	Mechanism
2	Study of animal behaviour	3	8 h	Synthesis	Mechanism
3	Species-specific handling techniques	2	3 h	Comprehension	Guided response
4	Animal health and disease prevention	3	7 h	Comprehension	Mechanism
5	Animal welfare and ethics	2	3 h	Evaluation	—

MODULE 6 · UNIT 1

Introduction to animal house management

3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

- Overview of animal house management
- Animal house design and layout
- Record-keeping and documentation

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe the principles of Animal House Management with its design and layout.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Design and Set Up an Animal House.	*design*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Present structure and functioning of an animal house.	*present*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 2

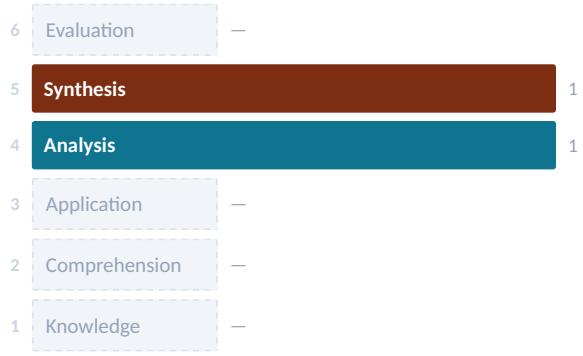
Study of animal behaviour

3 objectives | 8 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

Introduction to Animal Behaviour	Types of Animal behaviour.

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Identify behavioral patterns and body language in animals.	*identify*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Observe and. Interpret Animal Body Language through demonstration.	*observe*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *observe* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior.	*develop*	Procedural	Experiential 5h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 3

Species-specific handling techniques

2 objectives | 3 hours | Lecture 1 · Practical 1 | Outcomes: CO 7, CO 8

Non-human primates

Aquatic animals

Other specialized species (e.g., birds, reptiles)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the handling techniques for various laboratory animal species, ensuring handler safety.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Species-Specific Handling Techniques for Laboratory Animals.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 4

Animal health and disease prevention

3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

Common diseases in laboratory animals
Vaccination and immunization programs
Parasite control and management

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe Common diseases in laboratory animals and their management.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate microscopic Examination of Faecal Samples for Parasites.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Prepare and present a parasite Control in Laboratory Animals.	*prepare*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 5

Animal welfare and ethics

2 objectives | 3 hours | Lecture 1 · Practical 1 | Outcomes: CO 7, CO 8

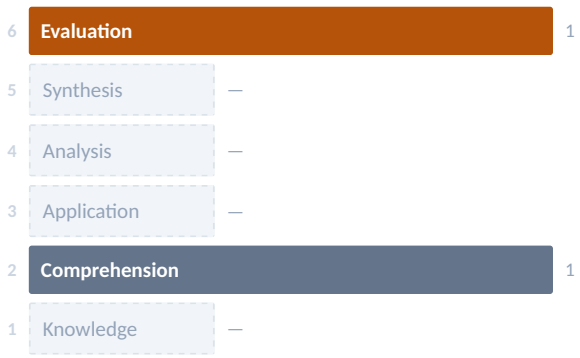
Animal welfare principles and ethics

Humane endpoints and euthanasia methods

Case studies and group discussions on animal welfare and ethics

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Animal Welfare Principles, Ethics, Humane Endpoints, Euthanasia Methods, and Case Studies in Animal Research.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Evaluate Animal Welfare and Ethics in a Research Setting.	*evaluate*	Conceptual	Practical 2h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7

Classical and Advanced quality control techniques for Ayurveda Formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	14	12	50
60 notional hours	26 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Classical and Advanced Quality Control Techniques for Ayurveda Formulations, provides a comprehensive understanding of quality control and quality assurance in Ayurveda formulations. It covers classical and advanced quality control techniques, including quality control parameters, regulatory requirements, and documentation. The module also explores quality control paramet

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

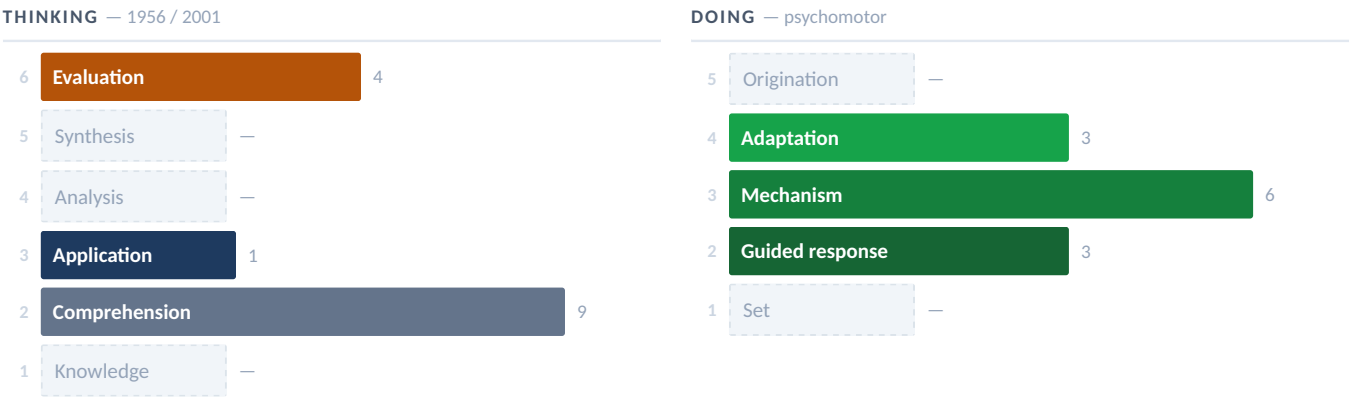
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	8	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	8	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the importance of quality control and quality assurance in Ayurveda product manufacturing.	*describe*	Level 2 Comprehension / Understand
2	Describe classical quality control parameters for Ayurvedic formulations, including Mana, Avaleha Siddhi Lakshan, and Bhasma Pariksha.	*describe*	Level 2 Comprehension / Understand
3	Apply quality control parameters for various Ayurvedic dosage forms, including solid, semi-solid, and liquid.	*apply*	Level 3 Application / Apply
4	Identify regulatory requirements for quality control and ensure documentation and record-keeping compliance.	*identify*	Level 1 Knowledge / Remember
5	Integrate classical and advanced quality control techniques to ensure the quality and safety of Ayurvedic formulations.	*integrate*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Quality Control and Quality Assurance	4	6 h	Comprehension	Adaptation
2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	6 h	Application	Mechanism
3	Quality Control parameter for solid dosage forms	9	26 h	Evaluation	Mechanism
4	Quality Control parameter for semi-solid dosage forms	3	6 h	Evaluation	Mechanism
5	Quality Control parameter for liquid and gaseous dosage forms	6	12 h	Evaluation	Adaptation

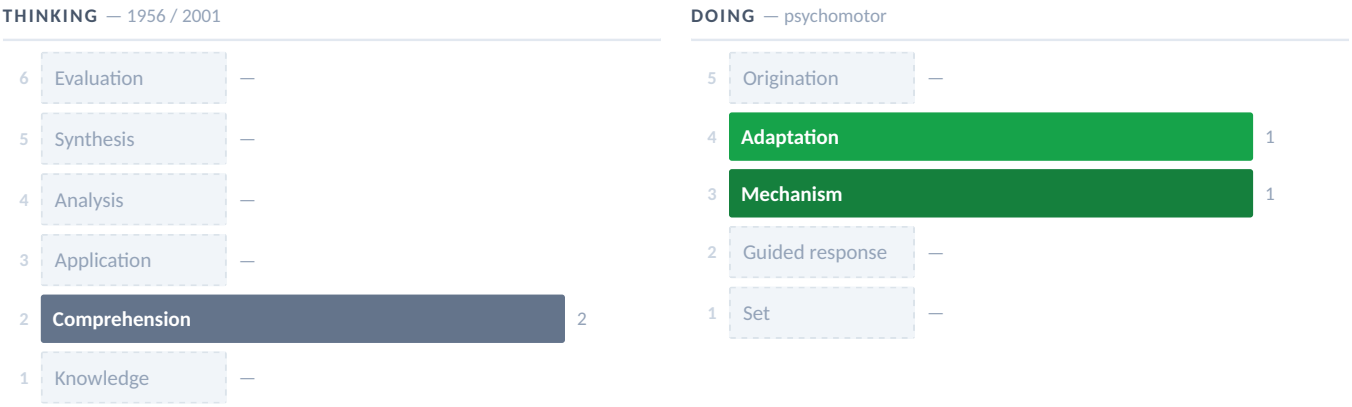
MODULE 7 · UNIT 1

Quality Control and Quality Assurance

4 objectives | 6 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 5

- Importance of quality control in Ayurveda
- Quality Assurance of Ayurveda Products
- Quality control parameters (identity, purity, etc.)
- Regulatory requirements for quality control
- Documentation and record-keeping

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the importance of Quality Control and Quality Assurance in Ayurveda.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality Control Parameters, Regulatory Requirements, and Documentation.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 3 · MECHANISM — 1				
Practice documentation and record-keeping processes in the quality control section.	*practice*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.
DOING — LEVEL 4 · ADAPTATION — 1				
Demonstrate documentation and record-keeping processes in the quality control section.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 2

Classical aspects of the quality control parameter for Ayurvedic formulations

4 objectives | 6 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 4

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Siddhi Lakshana of Avaleha, Snehapaka, Sandhana and Guggulu Paka.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss Mana and their application in quality control and Bhasma/Pishti pariksha.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	*assess*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 3

Quality Control parameter for solid dosage forms

9 objectives | 26 hours | Lecture 3 · Practical 3 · Experiential 3 | Outcomes: CO 4

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3				
Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for solid dosage forms like tablets, pills, and Varti.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 2				
Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*demonstrate*	Procedural	Practical 3h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
Demonstrate quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	*demonstrate*	Procedural	Practical 4h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 3				
Perform Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Perform quality control procedures for solid dosage forms.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
Perform quality control procedures for Rasoushadhi.	*perform*	Procedural	Experiential 5h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 4

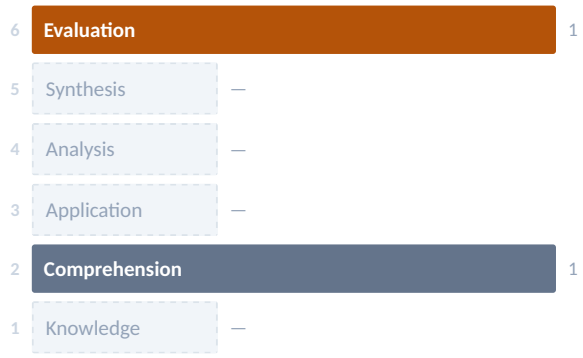
Quality Control parameter for semi-solid dosage forms

3 objectives | 6 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 4

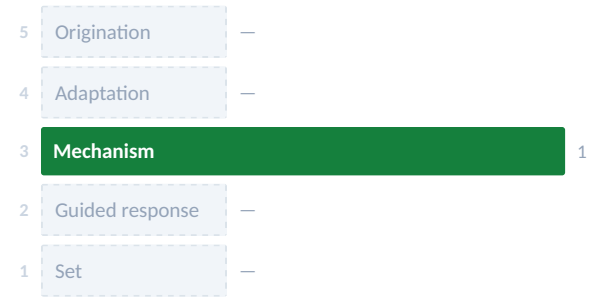
Rasakriya/Avaleha/Paka	Anjana

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	*describe*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Demonstration of Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	*demonstrate*	Procedural	Practical 2h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Perform quality control tests for semisolid dosage forms.	*perform*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

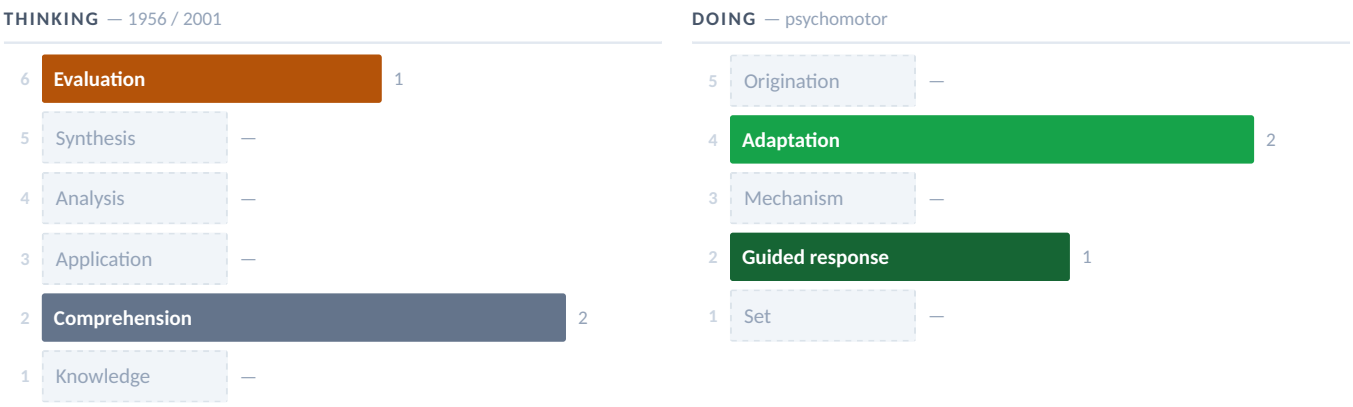
MODULE 7 · UNIT 5

Quality Control parameter for liquid and gaseous dosage forms

6 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 2 | Outcomes: CO 1, CO 4

Kashaya	Asava-arishta
Arka	Netra Bindu/Karna Bindu
Sharkara	Aerosols/sprays

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss Quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Demonstrate quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	*demonstrate*	Procedural	Practical 2h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate quality control parameters for Netra Bindu/ Karna Bindu and aerosols/sprays.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 2				
Perform quality control tests for liquid dosage forms.	*perform*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.
Perform quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays.	*perform*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8

Nano-science, nano-technology and its application in Ayurveda

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	11	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Understanding Nano-Science, Nano-Technology, and its Application in 2 10 20 30 60 Ayurveda covers integration of nano-science and nano-technology withAyurveda, exploring its potential to revolutionize the field of traditional medicine. This module explores the fundamentals of nanotechnology, its applications in Ayurvedic formulations, and the regulatory and safety aspects of this emerg

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

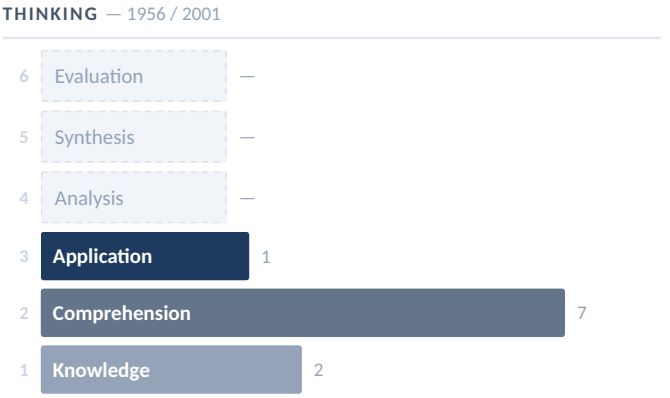
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

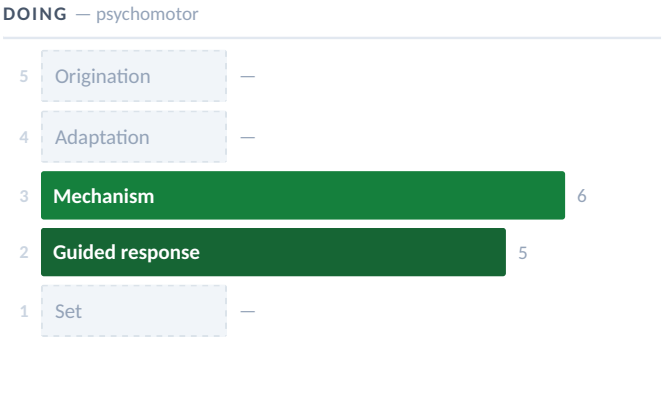
KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.





MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Define nanotechnology, its history, principles, and applications.	*define*	Level 1 Knowledge / Remember

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe Ayurveda Formulations with respect to Nanotechnology	*describe*	Level 2 Comprehension / Understand
3	Discuss the potential of nanotechnology in Ayurvedic formulations.	*discuss*	Level 2 Comprehension / Understand
4	Identify and describe various types of nanocarriers for Ayurvedic formulations.	*identify*	Level 1 Knowledge / Remember
5	Discuss the regulatory frameworks, safety concerns, and toxicity issues related to nanotechnology-based Ayurvedic formulations.	*discuss*	Level 2 Comprehension / Understand
6	Describe recent advances and future directions in nanotechnology-based Ayurvedic formulations.	*describe*	Level 2 Comprehension / Understand

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Nanotechnology	3	6 h	Application	Guided response
2	Ayurvedic Formulations and Nanotechnology	10	34 h	Comprehension	Mechanism
3	Nanocarriers for Ayurvedic Formulations	3	6 h	Comprehension	Guided response
4	Regulatory Aspects and Safety Concerns	2	2 h	Comprehension	—
5	Recent Advances and Future Directions	3	8 h	Comprehension	Mechanism

MODULE 8 · UNIT 1

Introduction to Nanotechnology

3 objectives | 6 hours | Lecture 2 · Practical 1 | Outcomes: CO 4, CO 8

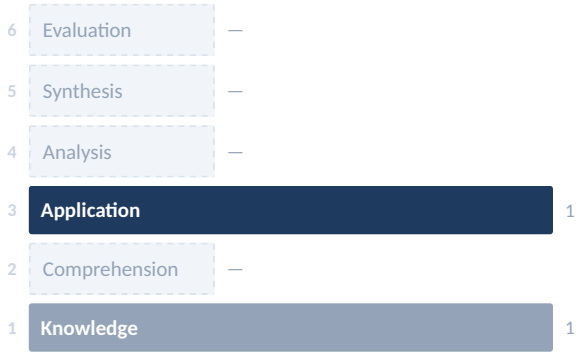
Definition and history of nanotechnology

Basic principles and applications of nanotechnology

Understanding Nanoscale

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Define nanotechnology and describe the history of nanotechnology.	*define*	Factual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the basic principles and applications of nanotechnology.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate the measurement of micro-particles using relevant equipment.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 2

Ayurvedic Formulations and Nanotechnology

10 objectives | 34 hours | Experiential 5 · Practical 3 · Lecture 2 | Outcomes: CO 4, CO 5, CO 8

- Overview of Ayurvedic formulations and their limitations
- Potential of nanotechnology in Ayurvedic formulations
- Characterization micro/nanoparticles in Bhasma
- Characterization micro/nanoparticles in Kupipakwa Rasayana and Pottali
- Characterization of micro-nano sized particles generated during Bhavana
- Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method and grasp the fundamental principles underlying nanoparticle synthesis
- Market Survey of Nano-Based Herbal Formulations

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss an overview of Ayurvedic Nano formulations and their limitations.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Identify the potential of nanotechnology in Ayurvedic formulations.	*identify*	Factual	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 3				
Demonstrate particle size of Bhasma prepared with different numbers of Puta.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Demonstrate particle size of various Kupipakwa Rasayana samples.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method.	*synthesize*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 5				
Measure particle size of Bhasma prepared with different numbers of Puta.	*measure*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Measure particle size of various Kupipakwa Rasayana samples.	*measure*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
Review articles regarding the green synthesis of metallic nanoparticles.	*review*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.
Analyze the impact of Bhavana by measuring the particle size of samples.	*analyze*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Conduct a market Survey of Nano-Based Ayurvedic Formulations.	*conduct*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 3

Nanocarriers for Ayurvedic Formulations

3 objectives | 6 hours | Lecture 2 · Practical 1 | Outcomes: CO 4, CO 5

Types of nanocarriers (liposomes, nanoparticles, etc.)

Advantages and limitations of nanocarriers in Ayurvedic formulations

Conversion of Taila into Oil in water nano emulsion and its characterization

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss types of nanocarriers (liposomes, nanoparticles, etc.).	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss the advantages and limitations of nanocarriers in Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate converting Taila into an oil-in-water nanoemulsion and its characterization.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
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MODULE 8 · UNIT 4

Regulatory Aspects and Safety Concerns

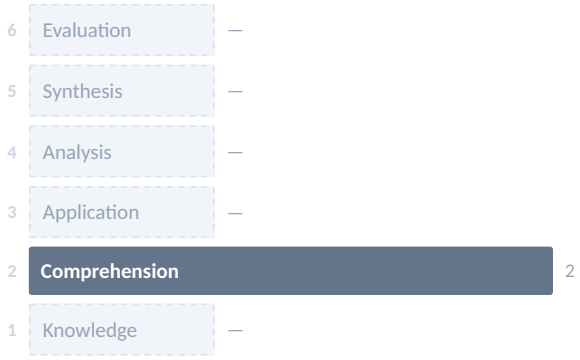
2 objectives | 2 hours | Lecture 2 | Outcomes: CO 5, CO 8

Regulatory frameworks for nanotechnology-based Ayurvedic formulations

Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 5

Recent Advances and Future Directions

3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 5, CO 8

> Recent advances in nanotechnology-based Ayurvedic formulations
> Future directions and potential applications
>
> Exploring the Future of Nanomedicine with Ayurvedic Formulations

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Discuss recent advances in nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss future directions and potential applications of nanotechnology in Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 3 · MECHANISM — 1				
Integrate nanomedicine with Ayurvedic formulations and their prospects.	*integrate*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

PART C

Reference

Nine appendices, all worked out from the curriculum's own objectives, so the counts here and the counts in Part B are the same numbers. C1 and C2 decode anything in the syllabus document. C3 compares every module and unit at a glance. C6 and C7 reproduce all ninety-one practical and experiential activities with the level each serves, so a reference like "Experiential Learning 8.2" can be looked up without turning back to the syllabus.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every module and unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the semester
6. **C6** All 47 practical training activities, mapped
7. **C7** All 44 experiential learning activities, mapped
8. **C8** Where the hours go, module by module
9. **C9** Sources and further reading

↑ Contents

PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column 3E of Table 3, with the domain it belongs to, the level it sits at, and how many objectives carry it. Keep this beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED THIS SEMESTER
CK	Thinking	1	Knowledge / Remember	8
CC	Thinking	2	Comprehension / Understand	55
CAP	Thinking	3	Application / Apply	15
CAN	Thinking	4	Analysis / Analyse	15
CS	Thinking	5	Synthesis / Create	4
CE	Thinking	6	Evaluation / Evaluate	8
PSY-SET	Doing	1	Set	4
PSY-GUD	Doing	2	Guided response	26
PSY-MEC	Doing	3	Mechanism	28
PSY-ADT	Doing	4	Adaptation	5
PSY-ORG	Doing	5	Origination	2
AFT-REC	Going about it	1	Receiving	not used
AFT-RES	Going about it	2	Responding	1
AFT-VAL	Going about it	3	Valuing	not used
AFT-SET	Going about it	4	Organisation	not used
AFT-CHR	Going about it	5	Characterisation	not used

Sixteen codes are defined; 12 are used here. Unlike the undergraduate curriculum, this semester **does** use Synthesis and Origination.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** — the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** — the schemes differ here	judges it against stated criteria

The top two levels are the only place the schemes disagree. This book follows the 1956 numbering, because that is what the curriculum prints in Column 3E.

PART C · APPENDIX C3

Level profile of every module and unit

All eight modules and forty-two units in one view. The fastest way to see where a semester's weight sits and to check that an assessment matches its module's ceiling.

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
MODULE 1 · PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY — 24 OBJECTIVES, 56 H, CEILING EVALUATION / ADAPTATION						
1	1	Introduction to Pharmaceutics	6	12	Evaluation	Adaptation
1	2	Pharmaceutical Dosage Forms	4	8	Analysis	Mechanism
1	3	Pharmaceutical Formulation	5	16	Application	Mechanism
1	4	Pharmaceutical Manufacturing and Quality Control	5	10	Evaluation	Mechanism
1	5	Biopharmaceutics	4	10	Application	Mechanism
MODULE 2 · DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH — 22 OBJECTIVES, 56 H, CEILING EVALUATION / MECHANISM						
2	1	Introduction to Databases and Information Systems	5	12	Application	Mechanism
2	2	Metal and Mineral Databases	6	16	Synthesis	—
2	3	Plant Databases	4	12	Analysis	—
2	4	Database application	4	12	Evaluation	Mechanism
2	5	Challenges and limitation of Databases	3	4	Analysis	—
MODULE 3 · ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES — 21 OBJECTIVES, 56 H, CEILING ANALYSIS / ORIGINATION						
3	1	Atomic, Molecular Structure, and Periodic Table Elements	3	12	Knowledge	Mechanism
3	2	Specialized Chemistry Fields-An Outline	2	6	Comprehension	—
3	3	Organic Chemistry	3	8	Analysis	—
3	4	Phytochemistry	7	16	Comprehension	Mechanism
3	5	Metallurgy	6	14	Comprehension	Origination
MODULE 4 · CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES — 15 OBJECTIVES, 56 H, CEILING SYNTHESIS / ADAPTATION						
4	1	Shodhana and Marana Procedures	3	3	Analysis	—
4	2	SOP of Shodhana Processes	4	22	Comprehension	Adaptation
4	3	SOP of Marana process	4	22	Comprehension	Mechanism
4	4	Chemical Changes involved in Shodhana and Marana processes	1	1	Comprehension	—
4	5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	8	Synthesis	—
MODULE 5 · PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS — 29 OBJECTIVES, 84 H, CEILING SYNTHESIS / MECHANISM						
5	1	Puta and its modern aspects	4	14	Synthesis	Guided response
5	2	Musha and its modern aspects	4	14	Analysis	Mechanism
5	3	Chromatography	5	17	Application	Mechanism
5	4	Spectroscopy	4	9	Analysis	Set

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
5	5	Microscopy	4	12	Comprehension	Mechanism
5	6	Calibration techniques	3	7	Analysis	Set
5	7	Overview of characterization instruments	5	11	Analysis	—
MODULE 6 · ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE — 13 OBJECTIVES, 28 H, CEILING EVALUATION / MECHANISM						
6	1	Introduction to animal house management	3	7	Comprehension	Mechanism
6	2	Study of animal behaviour	3	8	Synthesis	Mechanism
6	3	Species-specific handling techniques	2	3	Comprehension	Guided response
6	4	Animal health and disease prevention	3	7	Comprehension	Mechanism
6	5	Animal welfare and ethics	2	3	Evaluation	—
MODULE 7 · CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS — 26 OBJECTIVES, 56 H, CEILING EVALUATION / ADAPTATION						
7	1	Quality Control and Quality Assurance	4	6	Comprehension	Adaptation
7	2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	6	Application	Mechanism
7	3	Quality Control parameter for solid dosage forms	9	26	Evaluation	Mechanism
7	4	Quality Control parameter for semi-solid dosage forms	3	6	Evaluation	Mechanism
7	5	Quality Control parameter for liquid and gaseous dosage forms	6	12	Evaluation	Adaptation
MODULE 8 · NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA — 21 OBJECTIVES, 56 H, CEILING APPLICATION / MECHANISM						
8	1	Introduction to Nanotechnology	3	6	Application	Guided response
8	2	Ayurvedic Formulations and Nanotechnology	10	34	Comprehension	Mechanism
8	3	Nanocarriers for Ayurvedic Formulations	3	6	Comprehension	Guided response
8	4	Regulatory Aspects and Safety Concerns	2	2	Comprehension	—
8	5	Recent Advances and Future Directions	3	8	Comprehension	Mechanism

Both ceilings are shown because they move independently. Module 3 reaches only Analysis on the thinking ladder but Origination on the doing ladder, the highest doing level in the semester.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 171 objectives, sorted by the level that verb belongs to. The verb is what a student meets, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	identify 8, define 3, enlist 1, observe 1
Level 2	**Comprehension** Understand	discuss 42, describe 26, classify 1, illustrate 1, present 1
Level 3	**Application** Apply	demonstrate 34, perform 11, apply 2, measure 2, prepare 1, conduct 1
Level 4	**Analysis** Analyse	analyse 10, explore 4, analyze 3, differentiate 2, search 1, review 1
Level 5	**Synthesis** Create	design 4, develop 3, formulate 2, synthesize 1, integrate 1
Level 6	**Evaluation** Evaluate	evaluate 2, assess 1
Affective	**Attitude verbs**	practice 1

Choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, Part B says which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the semester

The second axis of the 2001 scheme, counted across all 171 objectives.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED
Factual	10	Rehearsal and self-testing. Only ten objectives here — this is not a memorisation course.
Conceptual	73	Explaining it aloud to somebody without notes.
Procedural	96	Carrying it out, more than once. **The dominant kind in this semester**, and the reason reading cannot substitute for the practical hours.
Metacognitive	not used	Being made to look back at how you worked. Not asked for here.

An objective working on two kinds is counted under both, so the total exceeds 171. The kind is read from the wording of each objective; where the wording does not settle it, the label carries a small circle (°).

PART C · APPENDIX C6

All 47 practical training activities, mapped

Table 4 of the curriculum, with the level each activity is cited to serve. Practical training is where a student moves from performing with help to performing unaided.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY			
1.1	Demonstration of regulatory frameworks (FDA,etc.)	Doing 1 · Set	The teacher will demonstrate a scenario where a pharmaceutical company is developing a new drug for a specific disease condition. He will identify and explain key regulations and guidelines the company must follow for IND (Investigational New Drug) application submission. Students are asked to explore specific regulatory guidelines applicable to different regulatory bodies, construct the process of application, and present
1.2	Demonstration of pharmaceutical product development process	Doing 2 · Guided response	The teacher will demonstrate a video presentation (15 to 25 minutes) on the pharmaceutical product development process, covering introduction to a selected drug candidate, discovery and target identification, preclinical testing and formulation development, clinical trials, regulatory submission and approval, and commercialization and post-marketing surveillance.
1.3	Identify and categorise Classical Ayurvedic dosage forms	Thinking 4 · Analysis	Each student will identify and classify different classical dosage forms mentioned in Ayurvedic classics into solid, liquid, and semi-solid categories under the guidance of the teacher. Students will record observations and characteristics of each dosage form.
1.4	Identification and categorisation of modified Ayurvedic dosage forms	Thinking 4 · Analysis	Each student will identify and classify different modified Ayurvedic dosage forms into solid, semi-solid, liquid, and gaseous categories under the guidance of the teacher. Students will examine and sort samples of different dosage forms, such as tablets, capsules, solutions, suspensions, ointments, creams, and infusions. Students will record observations and characteristics of each dosage form.
1.5	Formulation of Solid Dosage Forms selecting appropriate excipients	Doing 2 · Guided response	The teacher will design and prepare a simple tablet formulation using a model drug and excipients. Measure and mix the ingredients, compress the tablets and evaluate their physical characteristics. Students will be instructed to prepare the tablet following the same steps.
1.6	Formulation of liquid Dosage Forms selecting appropriate excipients	Doing 2 · Guided response	The teacher will prepare a simple syrup formulation using a model drug and excipients. Measure and mix the ingredients, dissolve the solids, and evaluate the physical and chemical characteristics of the syrup. Students are instructed to prepare a syrup following the same steps.
1.7	GMP audit and compliance evaluation	Thinking 6 · Evaluation	The teacher will explain the importance of GMP and cGMP in drug manufacturing. He will provide the checklist to conduct an audit. Under his guidance students will conduct a mock audit to evaluate compliance with Good Manufacturing Practices (GMP) and current Good Manufacturing Practices (cGMP) in a pharmaceutical manufacturing setting. Assess documentation, personnel training, equipment maintenance, and facility cleanliness. In the case of non-GMP certified teaching pharmacy, the student can conduct a mock audit and note down the observations comparing with checklist and present his findings mentioning the shortcomings if any, and required components to match GMP standards

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
1.8	Demonstration, designing, and evaluation of pharmaceutical packaging and labelling	Doing 2 · Guided response	The teacher will demonstrate a standard packaging and labeling design with a suitable example. He will then assign each student the task of designing and evaluating a pharmaceutical packaging and labeling system for a specific product, considering factors such as container closure systems, labeling requirements, and tamper-evident features. The teacher will assess the system's compliance with regulatory requirements and industry standards.
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH			
2.1	Demonstration of various searching techniques for databases	Doing 2 · Guided response	The teacher will demonstrate selected searching techniques for databases and demonstrate. Students will be informed to follow the techniques using various MeSH(Medical Subject Headings) terms
2.2	Demonstration of Modelling, Querying, and Analytics of Data	Doing 2 · Guided response	Teacher will Create a data model for a metal, mineral, or plant-related database. Students will be asked to write complex SQL (Structured Querry Language) queries to retrieve and analyze data. Perform data analytics tasks such as data visualization, data mining, and predictive modelling
2.3	Demonstration of techniques to search and extract data from the USGS Mineral Database	Thinking 4 · Analysis	The teacher will show how to search and extract data from the USGS Mineral Database on a specific mineral or metal. Students will be asked to follow the technique and extract the data for the assigned mineral. They are then asked to analyze and present the data in a report.
2.4	Analyse and integrate Web mineral Database	Thinking 5 · Synthesis	The teacher will demonstrate searching data for minerals from multiple databases and show how to integrate data from multiple metal and mineral databases to solve a real-world problem. Students will be advised to follow the technique to present the integrated database and discuss the challenges and limitations.
2.5	Search and explore Metal Database and Applications	Thinking 3 · Application	The teacher will demonstrate the technique to search and explore metal databases, including their applications and significance. Students will be advised to follow the technique and search metal databases and present a case study on the use of metal databases in industry or research
2.6	Demonstration of techniques to explore	Thinking 4 · Analysis	The teacher will demonstrate the techniques to search and extract data from The Plant List (TPL) and International Plant Names Index (IPNI) on a specific plant species or family. Students will be instructed to follow the technique, gather the data, analyze Plant Databases
2.7	Analysis of case studies of Database Applications in Metal, Mineral, or Plant Research	Thinking 6 · Evaluation	The teacher will demonstrate selected case studies of database applications in metal, mineral, or plant research and will explain how to evaluate the effectiveness of the database applications, identify best practices, and discuss potential improvements and future directions.
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES			
3.1	Analysis of chemical bonding types, principles, and isomerism	Doing 2 · Guided response	The teacher will assign students video resources on chemical bonding and isomerism. Students will watch videos and take notes. They will be asked to present their notes on a basic understanding of the relevant chemistry concept with an interpretation relating to the Shodhana and Marana process.
3.2	Demonstration of the essential fundamental principles of crystallography and organometalli	Thinking 2 · Comprehension	—
3.3	Demonstration of different types of extraction methods including solvent selection	Doing 2 · Guided response	The teacher will select an herb and a formulation and demonstrate extraction following different methods and solvents. The teacher will explain the process of recording the percentage yield and physical status of the respective extracts. Different herbs and formulations will be selected for each session to avoid repetition.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.4	Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations	Doing 2 · Guided response	The teacher will select an herb and a formulation and extract the same in specific solvents. Extracts will be tested for phytoconstituents by Test Tube methods. Different herbs and formulations will be selected for each session to avoid repetition.
3.5	Demonstration of different types of metals and their properties. The teacher will select d	Doing 2 · Guided response	—
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES			
4.1	Demonstration of the Shodhana procedure of Rasadravyas using various methods	Doing 2 · Guided response	The teacher will demonstrate a set of SOPs of Various Shodhana procedures like Nimajjana (Immersion), Bhavana (Levegation), Swedana (heating under a liquid bath), Pachana(Boiling),Patana(Sublimation/ distillation),Nirvapa(Quenching),Dhalana (Melting and pouring), Bharjana(Roasting/frying), Nirjaleekarana(Dehydration) etc. with a selected suitable drug, explain scientifically applying different theories, and discuss the rationale behind the procedure and Shodhana Dravya used
4.2	Demonstration of the Marana procedure of Rasadravyas with different methods	Doing 2 · Guided response	The teacher will demonstrate any three Marana procedures along with SOPs for each procedure with a selected suitable drug and type of Puta, analyzing each step of the Marana scientifically applying different theories and discussing the rationale behind Marana Dravya used in the procedure.
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS			
5.1	Demonstration of selected Puta to give the knowledge of Standard Operative Procedure	Doing 2 · Guided response	The teacher will select any one Puta and one type of fuel and demonstrate the Standard Operative Procedure of the selected Puta and explain the recording and documentation of the temperature pattern, and time frame. Different Puta/fuel should be selected for every batch to avoid repetition of the Puta/fuel.
5.2	Demonstration of the handling of Musha following the Standard Operating Procedure (SOP). T	Doing 2 · Guided response	—
5.3	Demonstration of the chromatography techniques to separate, The teacher will demonstrate t	Doing 1 · Set	—
5.4	Demonstration of spectroscopic instruments through Audiovisual aids	Doing 1 · Set	The teacher will show Audio-Visual Content related to the instrumentation and functioning of various spectroscopy instruments. After the demonstrations, the students will discuss the key points they observed. The teacher will summarize the contents. Further students are advised to observe the working of these instruments during their visit to the characterization lab as a part of module 8
5.5	Demonstration of handling of petrological and trinocular microscopes	Doing 2 · Guided response	The teacher will select any mineral and demonstrate the method of identifying rocks/minerals based on optical characters. For every batch different sample will be selected for demonstration (avoiding repetition).The teacher will select a suitable sample and demonstrate the microscopic characteristics of the sample like morphology and particle size with the help of projection and discuss the findings. For every batch different sample will be selected for demonstration (avoiding repetition)
5.6	Demonstration of calibration of various instruments present in the department	Doing 1 · Set	The concerned expert will demonstrate the methods of calibration of various equipment present in the department.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.7	Demonstration of characterization instruments with Audiovisual aids	Thinking 3 · Application	Each student will be instructed to search and collect articles related to specific instruments used for the characterization of Ayurvedic formulations like, PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions)
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE			
6.1	Demonstration of animal house model	Doing 2 · Guided response	The teacher will demonstrate an animal house model and describe its compartments considering space requirements, ventilation, temperature control, lighting, noise levels, sanitation, and hygiene including record-keeping and documentation.
6.2	Demonstration of Animal Body Language	Thinking 5 · Synthesis	Teacher will assign each student a small laboratory animal group. They will be provided a checklist of behavioural patterns and body language cues to observe, including posture, facial expressions, and vocalizations. They will be instructed to observe the animals for 30 minutes, recording their observations on the checklist. After the observation period, students will discuss and interpret their findings. Encourage students to present their findings to the class, facilitating a discussion on the importance of recognizing animal behaviour and body language. Note: Before visiting virtual observation using Computer Assisted Learning using dedicated soft ware – Ex Pharm may be carried out
6.3	Demonstration of Species-Specific Handling Techniques for Laboratory Animals	Doing 2 · Guided response	Under the guidance of experienced instructors, students will engage in hands-on handling sessions with various small laboratory animal species, including rodents, rabbits, and guinea pigs. Students will learn and practice species-specific handling techniques, focusing on animal behavior, body language, and welfare considerations. The session will conclude with a debriefing discussion, addressing challenges, best practices, and future applications.
6.4	Demonstration of microscopic Examination of Faecal Samples for Parasites	Doing 2 · Guided response	Under the supervision of the teacher or animal house in-charge, students will collect and label fecal samples from laboratory animals, wearing gloves and lab coats. They will prepare slides with staining solution and examine them under the microscope, starting at 40x and increasing to 100x. They will record parasite presence, morphology, and size. They will compare their findings with reference materials and participate in a debriefing discussion
6.5	Evaluation of Animal Welfare and Ethics in a Research Setting. The teacher will prepare ca	Thinking 6 · Evaluation	—
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS			
7.1	Demonstration of documentation and record-keeping processes in the quality control section	Doing 4 · Adaptation	The teacher will take the students to the quality control section and demonstrate the documentation and record-keeping processes for different dosage forms, from input materials to finished products.
7.2	Demonstration of assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	Doing 2 · Guided response	The teacher will select at least one Bhasma from each category (ore/ mineral, metal, and Sudha Varga) and demonstrate classical Bhasma Pareeksha. Additionally, special Bhasma Pareeksha tests for Rasadravyas (such as Amla Pareeksha for copper-containing compounds) will be performed. The scientific basis of these tests will be discussed. To illustrate the application of each test, the teacher will demonstrate Bhasma Pareeksha using both positive and negative samples.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.3	Demonstration of Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of powder dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.4	Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti	Doing 2 · Guided response	The teacher will select at least one formulation from each category of solid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.5	Demonstration of quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana, Lauha/ Mandura	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of Rasoushadhi (Chaturvidha Rasayana, Louha/Mandura) and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.6	Demonstrate Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/ Paka/Anjana as per	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of semisolid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia. Pharmacopeial standards
7.7	Demonstration of quality control parameters for liquid dosage forms like Kashaya, Arka, Sharakara, Asava-arishta	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of liquid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.8	Demonstration of quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays	Doing 2 · Guided response	The teacher will select at least one formulation from aerosols/sprays and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA			
8.1	Demonstration of the measurement of micro-particles using relevant equipment	Doing 2 · Guided response	The teacher will demonstrate the application of microscale measurement using relevant equipment. Various samples of differing dimensions, ranging from micro to centimeter levels, will be selected and measured using equipment like micrometers and vernier calipers. Under the teacher's guidance, students will identify microparticles from the selected samples.
8.2	Demonstration of particle size of Bhasma prepared with different numbers of Puta	Doing 2 · Guided response	The teacher will select Bhasma samples prepared with varying numbers of Puta and demonstrate the particle size measurement in these samples using a relevant microscope.
8.3	Demonstration of particle size of various Kupipakwa Rasayana samples	Doing 2 · Guided response	The teacher will select Kupipakwa Rasayana samples and demonstrate the measurable particle size in the samples using a relevant microscope.
8.4	Synthesiis of silver nano particles (AgNPs) using the chemical precipitatio n method	Doing 2 · Guided response	The teacher will demonstrate the synthesis of silver nanoparticles using the chemical method in the laboratory, utilizing silver nitrate solution, and compare the results with those of Rajata Bhasma
8.5	Conversion of Taila into oil-in-water nanoemulsion and its characterization. The teacher w	Doing 2 · Guided response	—

47 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C7

All 44 experiential learning activities, mapped

Table 5 of the curriculum. These are the largest block of hours in the semester and where the upper levels are built; the modular assessment is drawn from them.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY			
1.1	Recognising applications suiting various regulatory	Doing 4 · Adaptation	The teacher will assign individual students a scenario where a pharmaceutical company is developing a new drug (different dosage forms) for different health conditions. Identify key regulations and guidelines the company must follow for IND (Investigational New Drug) application submission. Review a case study of a company receiving an FDA warning letter for GMP violations and frameworks propose corrective actions. Prepare a short presentation summarizing findings and recommendations.
1.2	Listing out challenges and solutions in the pharmaceutical product development process	Thinking 6 · Evaluation	Each student will be assigned to identify challenges in pharmaceutical product development of different dosage forms, such as poor solubility and regulatory compliance issues. Design potential solutions and analyze case studies of companies that overcame similar challenges. Present solutions and receive peer feedback, then reflect on key takeaways and implications for real-world scenarios
1.3	Exploration, identification and collection of various classical and modified pharmaceutical dosage forms	Doing 3 · Mechanism	Each student will be informed to visit a pharmacy or hospital dispensary. The department obtains prior permission from the pharmacy or head of the institution as applicable. In the case of a hospital dispensary Students are made in batches depending on the size of the dispensary and post. Students are sent one by one and asked to collect samples from specific dosage forms. Search and collect samples of different dosage forms, such as tablets, capsules, solutions, suspensions, ointments, creams, injections, and infusions. Samples manufactured by different pharmacies may be collected and compared. Record the name, category, and characteristics of each dosage form. Return to the classroom and create a display or presentation showcasing the collected dosage forms
1.4	Selection of an Appropriate Excipient According to Solid and Liquid Dosage Form	Doing 3 · Mechanism	Students will perform an experiential exercise to select and justify appropriate excipients for solid and liquid dosage forms. They will choose an active pharmaceutical ingredient (API) and a specific dosage form, and then determine the required characteristics of the dosage form. Students will then select an excipient to help achieve the desired characteristics and justify their selection. This exercise will help students understand the role of excipients in pharmaceutical formulation and apply theoretical knowledge to practical problems. By completing this exercise, students will develop their critical thinking and problem-solving skills in pharmaceutical formulation.
1.5	Designing and development of Pharmaceutical Packaging systems and labelling	Doing 3 · Mechanism	The teacher will assign each student to design and develop a pharmaceutical packaging system for a specific product, considering factors such as container closure systems, labelling requirements (based on dosage forms and ingredients), and tamper-evident features. Students will be asked to review the labels of various dosage forms available in the dispensary, evaluate them, and present their observations

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
1.6	Optimization of Pharmaceutical Formulation for Enhanced Bioavailability	Doing 3 · Mechanism	The teacher will allot each student a poorly soluble drug and students will be instructed to design a novel pharmaceutical formulation to enhance its bioavailability. Students will conduct literature research, formulate hypotheses, design experiments, and test the formulation. Evaluate the results using pharmacokinetic parameters and present findings in a written report and oral presentation Modular Assessment Assessment method Hour IInstructions—Conduct a structured Modular assessment. The assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Create a database of bioequivalence and bioavailability studies for different drug products. 2. Select a pharmaceutical product and research its stability and storage requirements. 3. Investigate the role of excipients in the bioavailability of a pharmaceutical product. 4. Develop a manufacturing process for the dosage form, including selecting equipment, developing a batch record, and evaluating process validation. 5. Provide each student with a sample pharmaceutical product that is missing some information. Discuss the importance of each missing label and how it could impact patient safety or compliance. OR Any practical in converted form can be taken for assessment.(25 Marks) and Any experiential learning method in converted form can be taken for assessment (25 marks) 4
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH			
2.1	Design and Implement a Simple Database for a Plant Nursery	Doing 3 · Mechanism	Students will be assigned to design a simple database for a plant nursery. Create three tables: Plant/metal/mineral, Customer, and Order. Define the relationships between the tables. Use a DBMS to create the database and insert sample data. Write SQL queries to retrieve and manipulate data. Perform basic operations like adding a new plant, updating customer addresses, and deleting an order.
2.2	Analyse Web mineral Database and Metal Database	Thinking 4 · Analysis	Each student will be assigned a different metal/mineral (from the groups mentioned in Rasashastra like, Maharasa, Uparasa, Sadharana Rasa, Sudhavarga, Loha etc)and asked to conduct an in-depth analysis of the Webmineral, Mineral Database, and research metal databases to identify trends, patterns, and correlations. Present their findings and insights.
2.3	Exploration and analysis of plant database	Thinking 4 · Analysis	Each student will be asked to carry out an independent research project using The Plant List (TPL) and the International Plant Names Index (IPNI) to investigate a specific plant-related question or hypothesis. Collect and analyze data, and present findings in a research paper or presentation.
2.4	Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research	Doing 3 · Mechanism	Each student will analyze different case studies of database applications in selected metal, mineral, or plant research. Students will evaluate the effectiveness of the database applications, identify best practices, present the same, and get feedback from peers.
2.5	Perform a role play related to the challenges and limitations of Participate in a team rol	Affective · Responding	—
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES			
3.1	Exploration of the Periodic Table: Chemical Properties of Main Group and Transition Metals	Doing 3 · Mechanism	—
3.2	Exploration and identification of different functional groups and discuss their influence	Thinking 4 · Analysis	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.3	Extraction Techniques in Herbal Formulations: A Hands-on Approach	Doing 3 · Mechanism	Each student will be provided with different samples of herbal and polyherbal formulations. They will then extract the samples using various extraction techniques and solvents. The yield will be recorded, and the physical status of the extracts will be noted. To ensure a diverse learning experience, different herbs and formulations will be selected for each session.
3.4	Identification of phytoconstituents in Herbs and Herbal Formulations: A Practical Approach	Doing 3 · Mechanism	The teacher will assign a specific herb and formulation to each student. Students will then extract the allotted herb using designated solvents. The resulting extracts will be tested for phytoconstituents using qualitative test tube methods. To ensure a diverse learning experience, different herbs and formulations will be selected for each session.
3.5	Test different types of metals for their properties. Each student will be provided with va	Doing 5 · Origination	—
3.6	Hands-on experience with basic metal	Doing 5 · Origination	Each student will be provided with various metal samples and will perform hands-on experiments to identify the metals using basic techniques. They will observe and record the physical properties and perform simple chemical tests to identify the metals. The identification techniques observations and results will be noted Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Investigate chemical bonding in Ayurvedic herb-drug interactions, Study isomerism in Ayurvedic compounds, and develop a chemical bonding-based model for Ayurvedic formulation design. 2. Ask the students to design and develop Ayurvedic formulations using the principles of atomic structure, periodic table, chemical bonding, and isomerism. 3. Conduct a cost-benefit analysis of the process of Satvapatana of iron ore, including the costs of extraction, purification, and conversion. 4. Research and present on the bioinorganic chemistry of metal ions used in Ayurvedic formulations, including their coordination chemistry and biological activity. OR Any practical in converted form can be taken for assessment.(25 marks) AND Any experiential learning method in converted form can be taken for assessment (25 marks) 4
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES			
4.1	Performing Shodhana of Rasadravyas with various methods	Doing 4 · Adaptation	Each Student will be allotted different drugs for each procedure to perform and experience the process. Students will present their observations, followed by critical appraisal/summarization by the teacher for comprehensive understanding. Each student may be allotted a set of processes, such as: 1. Swedana, Dhalana, and Nirvapa 2. Bhavana 3. Bharjana and Patana 4. Nimajjana, Pachana, and Nirjaleekarana 5. Bhavana, Swedana, and Nirvapa 6. Dhalana, Bharjana, and Nirjaleekarana 7. Nimajjana, Pachana, and Patana
4.2	Performing Marana procedure of Rasadravyas with different methods	Doing 3 · Mechanism	Each student will be allotted different drugs and asked to carry out Marana as per the SOP laid down in classical texts/published literature and experience the procedure. Each student will present their observations. This will be followed by critical appraisal/summarisation by the teacher for a comprehensive understanding

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
4.3	Analysis of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies	Thinking 5 · Synthesis	Each student will be allotted different articles published in peer-reviewed journals on Shodhana and Marana procedures and asked to analyze the contents and present their observations. Presentation of observation by each student will be followed by critical appraisal/ summarisation by the teacher for a comprehensive understanding Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Develop a detailed chart comparing the effectiveness of various media used in the Shodhana process of Rasa Dravyas, including their chemical composition, pharmacological properties, and therapeutic application. 2. Make a study on the effect of temperature and pressure on Shodhana and Marana, including the rate of reaction and the yield of product. 3. Conduct a comparative analysis of Shodhana and Marana, highlighting their similarities and differences at the physical and chemical levels. OR Any practical in converted form can be taken for assessment(25 Marks) AND 4 Any experiential learning method in converted form can be taken for assessment (25 marks)
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS			
5.1	Developing Standard Operative Procedures (SOPs)for given Puta	Thinking 5 · Synthesis	The teacher will allot any one Puta and one type of fuel to each student to develop the Standard Operative Procedure of the given Puta and the student will gain experiential learning by carrying out this practical and documenting the temperature required, and time frame. Different Puta/fuel will be allotted for each student (repetition of the Puta/fuel should be avoided in the next session)
5.2	Developing Standard Operative Procedure for handling Musha	Doing 3 · Mechanism	The teacher will allot a different drug to each student to determine its ash value using a Musha/ following a Standard Operative Procedure. The student will gain experiential learning by carrying out this practical and documenting the temperature required, time frame, and confirmatory test. Different drugs will be allotted for each student (repetition of the drug should be avoided in the next session)
5.3	Performance of chromatography to separate, and identify components and interpretation of r	Doing 3 · Mechanism	—
5.4	Analysis of published articles on spectroscopic instruments	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to specific spectroscopic instruments used for the characterization of Ayurvedic formulations. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions)
5.5	Hands-on practice on the trinocular microscope	Doing 3 · Mechanism	The teacher will allot separate samples to each student and the student will examine the given sample using the trinocular microscope and experience the microscopic characteristics of the given sample like morphology and particle size with the help of projection and discuss the findings.
5.6	Analysis of articles related to challenges in the calibration of equipment	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to the challenges in the calibration of equipment. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions). The teacher will encourage the students to develop a Standard Operative Procedure for the calibration of classical instruments

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.7	Analysis of published articles on advanced instruments used for characterization	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to specific instruments used for the characterization of Ayurvedic formulations like, PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions) Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 75 marks. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. 1. Design a quality control protocol for Ayurvedic formulation using chromatography. 20 Marks AND 6 Make a chart on the dimensions of putas and temperature given for Rasa dravyas. - 20 Marks OR Approach to conduct a comparative study of different analytical methods for standardization of Ayurvedic medicine.- 20 Marks AND Investigate the application of calibration techniques in emerging technologies, such as nanotechnology. - 20 marks 2. Create a presentation that explores advanced microscopy techniques, such as super-resolution microscopy and single-molecule microscopy. 35 Marks Or 1. Any practical in converted form can be taken for assessment.(40 Marks) AND 2. Any experiential learning method in converted form can be taken for assessment (35 marks)
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE			
6.1	Visit to an Animal House	Doing 3 · Mechanism	Students will visit a local animal house, with prior permission from the concerned authority, to critically observe and thoughtfully learn about animal house management practices. Beforehand, they will research and review relevant principles, prepare insightful questions, and create a comprehensive observation checklist. During the visit, students will carefully observe animal enclosures, feeding, and care practices. They will interact collaboratively with staff, take detailed notes and photographs, and reflect deeply on their observations and learning. After the visit, students will discuss their experiences empathetically with peers and the facilitator. They will then develop a well-informed action plan for improving animal house management practices.
6.2	Analysis of online resources to develop critical thinking and research skills, and the ana	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
6.3	Preparation and presentation of a parasite Control in Laboratory Animals	Doing 3 · Mechanism	Students will be instructed to search video programs that provide an overview of parasite control in laboratory animals, covering common parasites, best practices, and preventions. Best practices for parasite control are then presented, including sanitation and hygiene protocols, fecal examination and testing, and treatment and prevention strategies. Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 25 marks. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. 25 marks may be allotted to any one of the following assessment methods: 1. Students are advised to select different animal species and conduct a study on the behaviour of a specific animal species, including their habitat, diet, and social structure. 2. Develop an SOP for laboratory animal care and management, including information on animal handling and restraint, housing and enrichment, and health monitoring. 3. Investigate a case of animal cruelty and analyze the factors that contributed to cruelty, make a critical evaluation on existing animal welfare policies, and recommend improvements. OR Any practical in converted form can be taken for assessment.(25 Marks) OR Any experiential learning method in converted form can be taken for assessment (25 marks) 2
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS			
7.1	Practice of documentation and record-keeping processes in the quality control section	Doing 3 · Mechanism	Each student will be allotted a different dosage form and tasked with developing documentation and record-keeping processes according to an established proforma. Students will then present their work and engage in peer discussion.
7.2	Assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	Doing 3 · Mechanism	The teacher will allot different samples of Bhasma from the categories of ore/mineral, metal, and Sudha Varga to each student. Students will then conduct Bhasma Pareeksha according to classical guidelines. To facilitate a comprehensive understanding, the teacher may provide both positive and negative samples for testing.
7.3	Performance of Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	Doing 3 · Mechanism	The teacher will allot different powder dosage form samples to each student. Students will then conduct quality testing on their assigned samples according to standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.4	Performance of quality control procedures for solid dosage forms	Doing 3 · Mechanism	The teacher will allot different solid dosage form samples like tablets, pills, and Varti to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.5	Performance of quality control procedures for Rasoushadhi	Doing 3 · Mechanism	The teacher will allot different samples of Rasoushadhi (representing Chaturvidha Rasayana, Loha, and Mandura categories) to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.6	Performance of quality control tests for semisolid dosage forms	Doing 3 · Mechanism	The teacher will select at least one formulation from each category of semisolid dosage form(like Rasakriya/Avaleha/Paka/Anjana) and will demonstrate the quality testing methods as per the parameters prescribed in the Pharmacopoeia. The students will then conduct quality testing on their assigned samples as per the standard pharmacopoeial protocol. The results obtained will be compared to pharmacopoeial standards and discussed.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.7	Performance of quality control tests for liquid dosage forms	Doing 4 · Adaptation	The teacher will allot different liquid dosage form samples like Kashaya, Arka, Sharakara, Asava-arishta to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.8	Performance of quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays. The	Doing 4 · Adaptation	—
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA			
8.1	Measurement of particle	Doing 3 · Mechanism	The teacher will allot different Bhasma samples prepared with different numbers of Puta and instruct students to measure the size size of Bhasma prepared with different numbers of Puta of particles measurable in the samples using the relevant microscope.
8.2	Measurement of particle size of various Kupipakwa Rasayana samples	Doing 3 · Mechanism	The teacher will allot different samples of Kupipakwa Rasayana to students and instruct them to measure the size of particles measurable in the samples using the relevant microscope
8.3	Review of articles regarding the green synthesis of metallic nanoparticles	Doing 3 · Mechanism	Students will be instructed to explore relevant articles related to the green synthesis of metallic nanoparticles, analyze the contents, and discuss with peer
8.4	Experiencing the impact of Bhavana	Doing 3 · Mechanism	The teacher will instruct students to collect samples prepared at different numbers/duration of Bhavana and they will be instructed to analyze the particle size of the samples.
8.5	Market Survey of Nano-Based Ayurvedic Formulations	Doing 3 · Mechanism	The teacher will instruct students to conduct a market survey of nano-based Ayurvedic formulations and present the summary of their observations in the departmental seminar
8.6	Integration of nanomedicine with Ayurvedic formulations and its prospects. The department	Doing 3 · Mechanism	—

44 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C8

Where the hours go, module by module

The lecture, practical and experiential split for each module, from Table 2. Read alongside Chapter 6.

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	TOTAL	MARKS
1	Pharmaceutics and Pharmaceutical Technology	2	10 h	20 h	30 h	60 h	50
2	Databases and Information Systems for Metal, Mineral, and Plant Research	2	10 h	20 h	30 h	60 h	50
3	Essential Chemistry :- Unlocking the Rasashastra Techniques	2	10 h	20 h	30 h	60 h	50
4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures	2	10 h	20 h	30 h	60 h	50
5	Principles, specification, SOPs of handling of classical and advanced instruments	3	15 h	30 h	45 h	90 h	75
6	Animal house and Animal handling technique	1	5 h	10 h	15 h	30 h	25
7	Classical and Advanced quality control techniques for Ayurveda Formulations	2	10 h	20 h	30 h	60 h	50
8	Nano-science, nano-technology and its application in Ayurveda	2	10 h	20 h	30 h	60 h	50
Semester		**16**	**80 h**	**160 h**	**240 h**	**480 h**	**400**

The split is fixed by the credit framework at one part lecture to two parts practical to three parts experiential. Experiential hours include the modular assessment, at two hours per credit. Two thirds of this semester happens with the teacher not talking.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies, the curriculum this book is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme Column 3E uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semester II — Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB).* New Delhi: NCISM. Applicable from the 2024–25 batch. Tables 2, 3, 4 and 5 are the source of every objective, code, hour, mode and activity in this book.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

Extracting this curriculum needed several repairs peculiar to its layout: domain codes split across line breaks and interleaved with the level beside them, hours joined to the end of an objective's own full stop, activity references torn apart mid-word, and the activity description tables which sit inside Table 3 and had to be separated from the objectives they follow. All 171 objectives, 91 activities and 480 notional hours reconcile against the curriculum's own credit framework.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, and marked with a small circle (°) where the wording supports a reading without settling it.